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WILDFIRE SPREAD PREDICTION AND LIVE DRONE-BASED VISUAL FEEDBACK SYSTEM

Project Report

Submitted by

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CERTIFICATE

This is to certify that the project titled **Wildfire Spread Prediction and live Drone-Based Visual Feedback System** is a record of the Bonafide work done by Spandan Kumar (*Reg. No. 220929337*) and H Anupam Nair (*Reg. No. 220929228*) submitted in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology (B.Tech.) in **MECHATRONICS,** School of Mechanical Engineering, Manipal Institute of Technology, Manipal, Karnataka, (A Constituent unit of Manipal Academy of Higher Education), during the academic year 2025-2026.

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ABSTRACT

The response to fire in a dense urban environment is determined by the speed of detection, verification and dispatch. The common chains of response require human reporting or verification by a fixed sensor. This project presents a simulated drone-based system for the detection and response of fire. The system is integrated with MATLAB for the publication of the suspected fire coordinate and the reception of the mission status, with an Arduino Nano for the implementation of the publisher-subscriber communication layer via the HC-05 Bluetooth module for the serial wireless exchange of data, and with the Unity game engine for the three-dimensional city environment, the autonomous navigation of the drone, the obstacle avoidance by the drone, the hybrid verification of the fire and the dispatch of the fire trucks. The drone searches for fires by flying over the area and searches for the FIRE coordinate, takes off into the air, reaches a constant flight height, flies through the city using a NavMeshAgent, avoids buildings, checks the fire using a trigger-zone and a raycast-target, and dispatches a response truck after verification of the fire. The system presents a complete cyber-physical emergency response system in a simulated environment, and it can be extended to real UAVs, to camera-based sensing, and to smart-city environments.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
UAV	Unmanned Aerial Vehicle
UGV	Unmanned Ground Vehicle
HC-05	Bluetooth Serial Communication Module
MATLAB	Matrix Laboratory
COM	Communication Port
CR/LF	Carriage Return and Line Feed
API	Application Programming Interface
IDE	Integrated Development Environment
LOS	Line of Sight

CHAPTER 1
INTRODUCTION

1.1 Introduction to the Area of Work

Fire incidents are among the most time-critical emergencies in urban and semi-urban environments. A small delay in identifying the correct location of a fire can result in rapid spread, loss of property, disruption of public infrastructure and risk to human life. The area of work for this project lies at the intersection of emergency response automation, unmanned aerial vehicle simulation, embedded communication and real-time three-dimensional visualization. Instead of treating fire response only as a dispatch problem, the project views the situation as a complete system in which detection, communication, navigation, verification and response are connected.

Unmanned aerial vehicles provide a useful approach for initial verification because they can travel quickly through space, observe a target from an elevated position and access locations that may be unsafe or difficult for human responders. In a city environment, a drone can act as an initial verification unit that reaches a reported location and supports decision-making before a full emergency response is deployed. This makes the system suitable for simulation-based study because the complete sequence can be tested without risk to real people or property.

The present work focuses on a simulated but integrated drone-based fire detection and response workflow. Unity is used to create the three-dimensional city and mission environment. MATLAB is used as the control and monitoring side for sending the suspected fire coordinate and receiving mission status. Arduino Nano and HC-05 Bluetooth modules are included to demonstrate an embedded publisher-subscriber communication circuit. The work therefore combines software simulation with an embedded communication concept.

The drone in the simulation is not manually flown throughout the mission. Once the coordinate is received, the drone performs a sequence of autonomous operations. It takes off from the drone pad, moves to a fixed flight height, travels toward the target, avoids obstacles using Unity navigation, checks the fire by a hybrid sensor method and supports the deployment of a fire truck. This gives the project a clear system-level workflow rather than a simple animation.

1.2 Present Day Scenario

Modern emergency response systems use a combination of human reporting, alarm systems, control rooms, dispatch networks and specialized vehicles. In many buildings, fire alarms and smoke sensors provide local detection. In open city areas, however, the first report may still depend on human observation. Even when an alarm is generated, the information available at the response center may be incomplete. Responders may know that an incident has been reported, but they may not have immediate confirmation about intensity, exact visibility, accessibility or surrounding obstacles.

Recent advances in Unmanned Aerial Vehicles (UAVs), in embedded systems as well as in simulation platforms make it possible to investigate more flexible responses to emergency situations. Aerial inspection, search and rescue, disaster scenarios, forest fires as well as situations that require situational awareness can be monitored using drones. Furthermore, simulation tools allow for the design and test of such systems before actually deploying hardware. Unity provides a practical environment for visualizing city layouts, moving agents and response actions. MATLAB provides a familiar engineering environment for data generation, communication testing and monitoring.

Bluetooth-based serial communication using HC-05 modules is simple compared with industrial wireless networks, but it is suitable for demonstrating the fundamental idea of wireless message transfer. Arduino Nano boards provide a compact embedded platform to implement a publisher-subscriber circuit for communication testing. The present project uses these tools together to build a prototype workflow that represents the larger idea of cyber-physical emergency response.

1.3 Motivation for the Project

The main motivation for the project is the need to reduce the delay between fire reporting, fire verification and response deployment. A reported fire coordinate alone may not be reliable enough to deploy a response vehicle in an optimized automated system. The system must verify whether the fire is actually present near the reported point. A drone is suitable for this verification step because it can move to the area, observe the target and transmit its result to a control system.

Another motivation is to create a project that is not limited to a single domain. A purely MATLAB project would not show a realistic mission environment, while a purely Unity project would not demonstrate external communication. A purely Arduino project would only show hardware communication without visual mission behavior. By combining MATLAB, Arduino, HC-05 and Unity, the project demonstrates a more complete system-level integration.

Previous work in fire detection and UAV monitoring often focuses on one part of the system, such as image-based fire detection, forest fire monitoring, or path planning. The present academic project does not claim to replace those methods, but it addresses the shortcoming of isolated demonstrations by combining coordinate communication, autonomous navigation, fire verification and response deployment into one workflow. The uniqueness of the adopted methodology is the integration of a hybrid sensor logic with a visual city simulation and embedded communication model.

The possible end result is significant because it can be used as a foundation for further development. The same architecture can later be extended with real UAV hardware, camera modules, GPS input, thermal sensors, Internet of Things networks and real-time control dashboards. In the current stage, the project provides a safe, low-cost and repeatable simulation platform for studying mission logic.

1.4 Scope of the Project

The scope of the project is limited to simulation and prototype communication. The drone, city, fire and fire truck are represented inside Unity. MATLAB sends a pre-defined or generated fire coordinate through serial communication and waits for the response status. The Arduino Nano and HC-05 communication setup is used to demonstrate publisher-subscriber style transfer of messages. The drone navigation is implemented with a NavMeshAgent and an elevated navigation plane so that the drone can move at flight height while avoiding building obstacles.

The fire detection is implemented using a hybrid logic. The fire object includes a trigger zone that detects whether the drone is close to the fire area. A separate raycast target is used to confirm line-of-sight detection. Fire is confirmed only when the drone is within the detection zone and the raycast is able to identify the target object. This reduces false confirmation caused by proximity alone.

The project does not include real flame sensing, thermal cameras, GPS-based outdoor navigation, real motor control, power supply design or full-size fire suppression hardware. These components are considered future extensions. The current work is designed to demonstrate mission architecture, communication logic and autonomous response in a controlled simulation environment.

1.5 Project Work Schedule

Tabel 1.1: Project Work Schedule

Phase	Work carried out	Outcome
Phase 1	Selection of project topic, study of UAV fire response concepts and identification of required tools	Project concept and system requirements were finalized
Phase 2	Preparation of Unity city environment, drone model, fire object, fire truck and drone pad	Initial simulation environment was created
Phase 3	Development of MATLAB serial message format and Arduino Nano with HC-05 publisher-subscriber circuit	Coordinate transfer and status communication were established
Phase 4	Implementation of Unity message receiver, serial bridge and mission controller	Unity was able to receive the coordinate and start the mission

Phase 5	NavMesh plane setup, obstacle configuration and autonomous drone movement	Drone navigation and obstacle avoidance were achieved
Phase 6	Hybrid trigger-zone and raycast-based fire detection and truck deployment	Fire confirmation and response sequence were completed
Phase 7	Testing, debugging, final demonstration and report preparation	Final working prototype and documentation were completed

1.6 Impact of the Work on Environment, Safety and Other Factors

The proposed drone-based fire detection and response system has a direct impact on safety, emergency response efficiency, and environmental protection. Fire incidents in urban areas can lead to loss of life, damage to property, air pollution, and disruption of public infrastructure. By using a drone as an initial verification unit, the system reduces the delay between fire reporting and fire confirmation. This improves the safety of people in the affected area because emergency action can be initiated earlier. The drone can inspect locations that may be unsafe or difficult for humans to approach, especially when the fire is located near buildings, roads, or restricted zones. In the simulated environment, the drone performs autonomous navigation, avoids obstacles, verifies the fire location, and triggers fire truck deployment. This reduces the dependency on manual inspection and represents a safer approach for early-stage emergency assessment.

Safety precautions were considered during both the hardware and software implementation of the project. On the hardware side, the Arduino Nano and HC-05 Bluetooth modules were connected using proper voltage levels and stable wiring to avoid loose connections, short circuits, or communication failure. Serial communication was tested carefully before integrating it with Unity so that incorrect data transmission would not affect the simulation. On the software side, the drone was controlled through a root object to prevent unstable movement of the visual model. NavMesh-based obstacle avoidance was used to ensure that the drone does not pass through buildings during movement. The fire detection system was also designed as a hybrid method using both a trigger zone and raycast confirmation. This prevents false fire confirmation and ensures that the drone confirms the fire only when it is close to the fire zone and has a valid line of sight to the fire target.

Environmental aspects were also considered in the concept of the project. Faster fire detection and response can reduce the spread of fire, which in turn reduces smoke generation, harmful gas release, and damage to surrounding structures. In real-world applications, such systems can help reduce the environmental impact caused by delayed fire response, especially in dense urban areas, industrial zones, and forest-edge settlements. Since the current project is simulation-based, it does not produce direct environmental pollution during testing. The use of Unity simulation also reduces the need for repeated physical trials, thereby saving material resources and lowering risk during the development stage. The project therefore supports the idea of safer and more sustainable emergency response through virtual prototyping and autonomous monitoring.

1.7 Uniqueness of the Methodology Adopted

The uniqueness of the methodology lies in the integration of multiple domains into a single working system. The project does not focus only on drone animation or only on fire detection. Instead, it combines MATLAB-based coordinate transmission, Arduino Nano and HC-05 based publisher-subscriber communication, Unity-based 3D simulation, NavMesh-based autonomous drone navigation, hybrid fire detection, and fire truck deployment into one complete workflow. This makes the system different from a simple simulation because each subsystem performs a specific role in the overall mission sequence.

Another unique aspect of the methodology is the hybrid fire verification technique. The system does not confirm fire only by reaching a coordinate. Instead, the drone checks the fire using a trigger zone and a raycast target. The trigger zone verifies that the drone has entered the fire detection region, while the raycast confirms that the fire target is actually visible. This improves the reliability of the detection process and reduces the possibility of false confirmation. The use of a separate fire raycast target also makes the system more flexible because the detection point can be adjusted according to the position and size of the fire object.

The project also uses a structured object arrangement in Unity, where the drone movement is controlled through a main root object and the visual drone model is placed as a child object. This separation helps avoid common simulation problems such as visual offset, incorrect rotation, unstable movement, and collider-related errors. Similarly, the fire truck deployment was designed with a landing point so that the drone can land on the truck after fire confirmation. These practical design decisions make the methodology stable, modular, and suitable for demonstration. Overall, the adopted methodology is unique because it presents a complete cyber-physical emergency response model in a simulated environment while still including real embedded communication through Arduino and HC-05 Bluetooth modules.

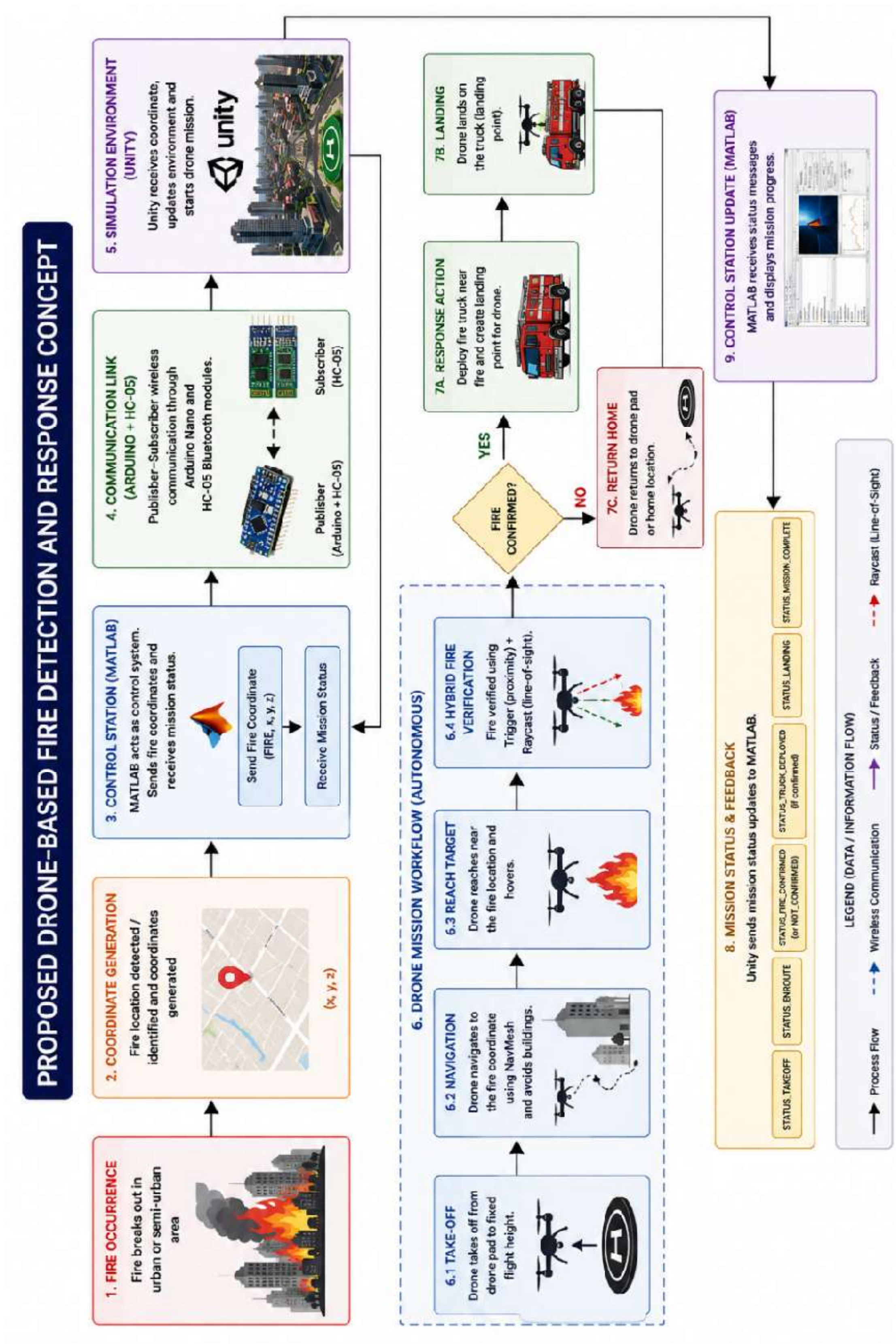

Figure 1.1: Proposed drone-based fire detection and response concept

CHAPTER 2
LITERATURE REVIEW AND OBJECTIVES

2.1 Literature Review

UAV-based monitoring has become an important area in disaster management because aerial platforms can provide fast spatial coverage and flexible viewpoints. Merino et al. presented an unmanned aircraft system for automatic forest fire monitoring and measurement, showing the role of UAVs in observing fire-affected regions [1]. Martinez-de Dios et al. discussed automatic forest-fire measuring using ground stations and unmanned aerial systems, indicating that UAVs can complement fixed sensing infrastructure [5]. Casbeer et al. studied cooperative forest fire surveillance using small UAV teams, which shows that aerial monitoring can be extended to collaborative multi-agent operations [18].

Video-based fire and smoke detection has also been widely studied. Yuan proposed a framework for video smoke detection using shape-invariant features and learning methods [2]. Celik et al. used statistical colour models for detecting fire in video sequences [3]. These works show that fire detection is often treated as a computer vision problem. However, in this project the focus is not deep image classification. Instead, a hybrid trigger and raycast method is used to demonstrate the principle of proximity and line-of-sight verification inside a real-time simulation.

Deep learning has recently improved UAV-based wildfire detection and segmentation. Ghali et al. reviewed deep learning and transformer approaches for UAV-based wildfire detection and segmentation [6]. Such methods can identify fire regions from aerial images, but they require datasets, computational resources and camera pipelines. The present project is at a prototype stage and therefore uses a simpler verification approach that can later be replaced with camera-based algorithms.

Path planning and obstacle avoidance are essential for autonomous agents. Hart et al. introduced the A* method, which became one of the fundamental graph-search approaches for determining efficient paths [11]. Karaman and Frazzoli reviewed sampling-based algorithms for optimal motion planning [4]. LaValle introduced rapidly exploring random trees as a tool for path planning [13]. Borenstein and Koren proposed the vector field histogram for real-time obstacle avoidance [12]. These works establish the importance of navigation algorithms in robotics. In this project, Unity NavMeshAgent is used as a practical implementation tool for path following and obstacle avoidance in a simulated environment.

Robotics and mobile robot literature also emphasizes the need for separating high-level mission planning from low-level movement control. Siciliano et al. describe core ideas in robotics modelling, planning and control [21]. Siegwart et al. explain mobile robot autonomy and navigation [24]. Craig provides the fundamentals of robot mechanics and control [23]. The project follows this separation by using MATLAB as the control and monitoring side, Unity scripts as mission logic, and NavMeshAgent as the movement execution component.

Embedded communication is another important part of the system. Arduino-based serial communication is widely used in academic prototypes because of its simplicity and availability. The Arduino Nano provides digital communication support in a compact board, and HC-05 Bluetooth modules can create a simple wireless serial link. In the present project, the Arduino Nano and HC-05 setup is used to represent a publisher-subscriber communication circuit, where one side publishes mission data and the other side receives and responds through status messages.

Game engines such as Unity are now commonly used for simulation, visualization and digital prototyping. Unity supports physics, transforms, navigation meshes, cameras, colliders, raycasts and serial integration through C# scripting. The ability to build a city scene, add a drone and control objects in real time makes Unity suitable for this project. The Unity Manual on NavMeshAgent, colliders and raycasts supports the implementation of the navigation and detection components [27], [28].

The literature suggests that UAV fire response systems require integration between sensing, communication, navigation and decision-making. Many papers emphasize one of these components separately. The current project contributes by constructing a combined academic prototype that connects coordinate input, embedded communication, autonomous drone travel, obstacle avoidance, hybrid verification and response deployment into one demonstrable workflow.

2.2 Summarized Outcome of Literature Review

The reviewed work shows that UAVs are useful for fire monitoring because they provide mobility, aerial perspective and flexible deployment. Vision-based detection methods are powerful but require image data and processing pipelines. Navigation research provides many methods for path planning and obstacle avoidance, but simulation environments are useful for demonstrating these concepts before hardware implementation. Embedded communication systems remain important because a practical cyber-physical system must transfer mission commands and status messages between control and execution units.

Based on the literature, the present project adopts a modular system. MATLAB is used for coordinate and status exchange, Arduino Nano and HC-05 are used for the publisher-subscriber communication circuit, and Unity is used for city visualization and drone mission control. The fire detection method is kept simple and explainable by combining trigger-zone proximity with raycast confirmation. This approach is suitable for an undergraduate project because it demonstrates the logic of reliable detection while remaining feasible to implement and test.

2.3 Objectives

The primary objective of the project is to design and implement a drone-based fire detection and response system using MATLAB, Arduino Nano, HC-05 Bluetooth modules, and Unity. The project aims to demonstrate a complete autonomous workflow in which a fire coordinate is transmitted, the drone navigates to the target, the fire is verified, and a response vehicle is deployed.

The design objective is to create a three-dimensional city environment with a drone, drone pad, fire object, fire detection zone, fire raycast target, fire truck, landing point, roads, and obstacles. The city environment must be suitable for demonstrating both navigation and response behavior. The drone must be organized using a root object and visual child model so that movement logic and visual representation remain stable.

The navigation objective is to implement autonomous drone movement using Unity NavMeshAgent. The drone must take off to a fixed height, travel toward the fire coordinate, avoid buildings, hover near the fire, and return or land depending on mission conditions. The obstacle avoidance objective is achieved through NavMesh obstacle configuration so that the drone does not pass through buildings.

The detection objective is to implement a hybrid fire detection system. The fire object includes a trigger zone to detect drone proximity and a separate raycast target to confirm line-of-sight detection. Fire is confirmed only when the drone is within the detection zone and the raycast successfully reaches the fire target. This objective reduces false confirmations and improves reliability.

The communication objective is to establish real-time data exchange between MATLAB and Unity. The project also implements a publisher-subscriber communication circuit using Arduino Nano and HC-05 Bluetooth modules. MATLAB publishes the fire coordinate, and Unity subscribes to the received coordinate and starts the drone mission. Unity then publishes mission status messages, which MATLAB receives and displays. This objective demonstrates an embedded communication layer in addition to software integration.

The response objective is to deploy a fire truck after fire confirmation. The truck must appear near the fire location, remain above the city ground, and provide a landing point for the drone. The system must also complete the mission by sending a mission-complete status.

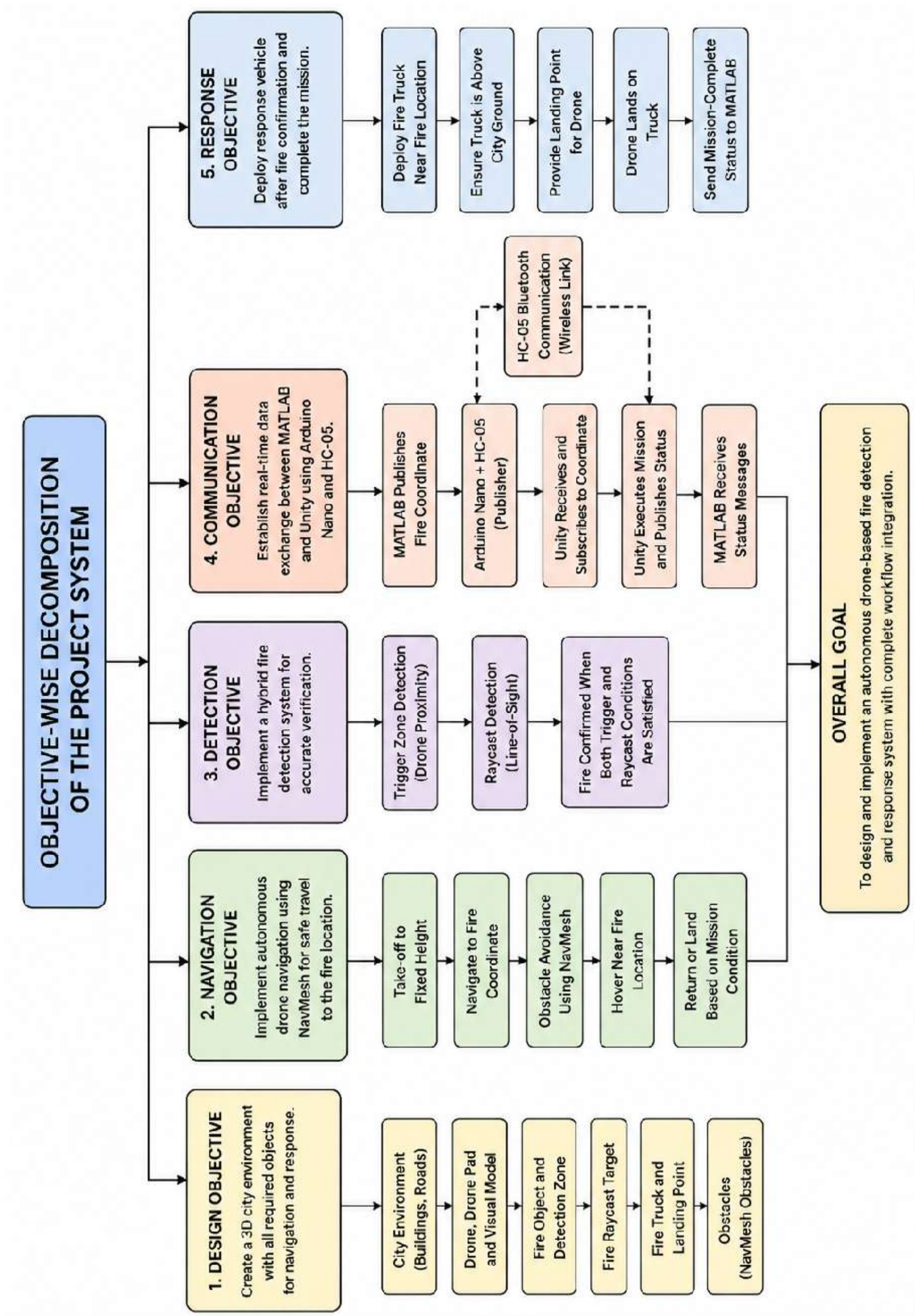

Figure 2.1: Objective-wise decomposition of the project system

CHAPTER 3
METHODOLOGY

The methodology adopted for this project follows a modular development approach in which the complete system is divided into communication, simulation, navigation, detection, and response modules. Each module was developed and tested individually before being integrated into the final working system. The project combines software simulation with embedded communication so that the fire coordinate can be transmitted, received, processed, and used to perform an autonomous mission inside the Unity environment. The overall methodology begins with MATLAB sending a fire location, followed by serial communication through the Arduino Nano and HC-05 Bluetooth module. Unity receives the coordinate, places or identifies the fire location, starts the drone mission, verifies the fire, deploys the fire truck, and finally sends mission status messages back to MATLAB.

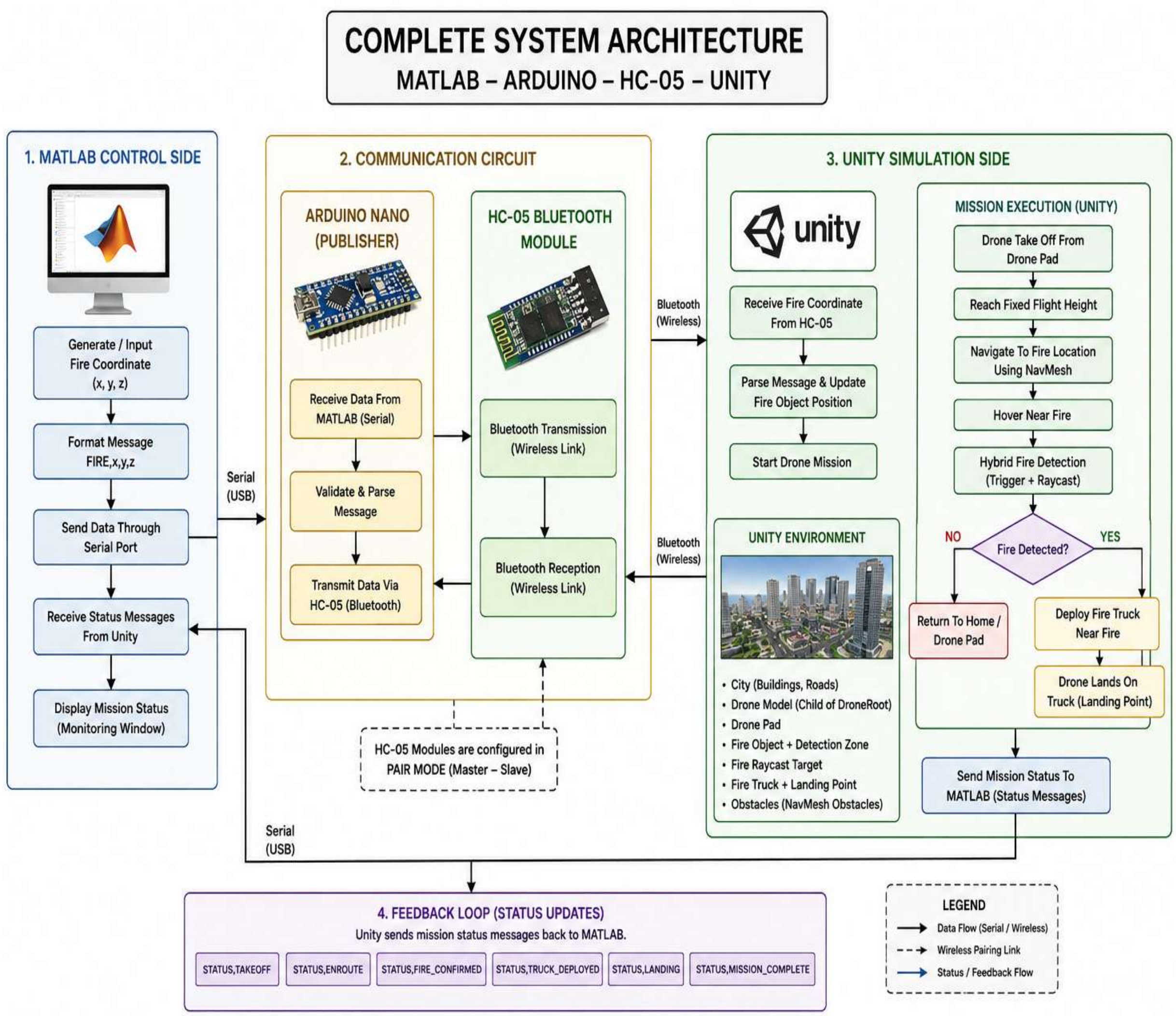

Figure 3.1: Complete system architecture with MATLAB, Arduino, HC-05, and Unity

3.1 Detailed Methodology

The first stage of the methodology was the creation of a three-dimensional city environment in Unity. The environment was designed to represent an urban area consisting of roads, buildings, a fire station, a drone pad, a fire location, a fire truck, and obstacles. The purpose of this environment was to provide a realistic simulation space where drone navigation, fire detection, and response deployment could be demonstrated. A separate drone navigation plane was created above the city to act as the movement layer for the drone. This plane was configured with Unity NavMesh so that the drone could move at a fixed height and avoid buildings or restricted areas during flight.

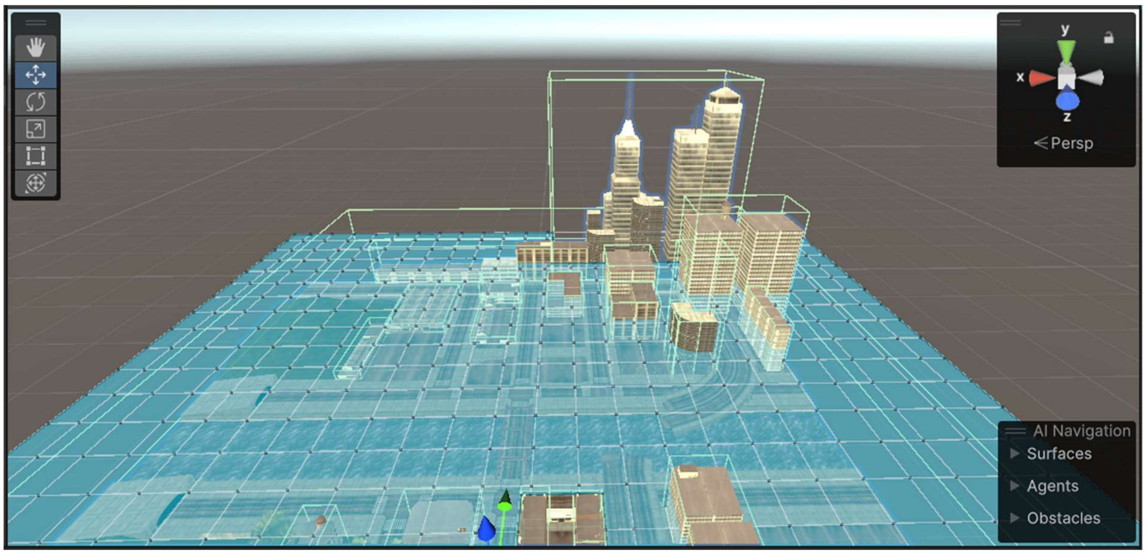

Figure 3.2: NavMesh flight layer and obstacle avoidance concept

The second stage involved setting up the drone system. The drone was arranged using a parent object called DroneRoot and a child object containing the visible drone model. The movement scripts and NavMeshAgent were attached to the root object, while the drone model was kept as a child for visual representation. This arrangement was used to avoid common movement problems such as incorrect rotation, unstable positioning, and visual offset. The drone was programmed to take off from the drone pad, move to the fixed flight height, travel toward the received fire coordinate, hover near the fire, and then either return or land on the fire truck depending on the result of fire verification.

The third stage involved implementing the communication system. MATLAB was used to send the fire coordinate in a structured message format. The coordinate message was transmitted through serial communication and received by Unity through the serial bridge script. The project also included a publisher-subscriber communication circuit using Arduino Nano and HC-05 Bluetooth modules. In this communication model, MATLAB acts as the publisher of the fire coordinate, while Unity acts as the subscriber that receives and responds to the coordinate. After the mission begins, Unity sends status updates such as fire confirmed, truck deployed, landing on truck, and mission complete. MATLAB receives and displays these status messages, completing the two-way communication process.

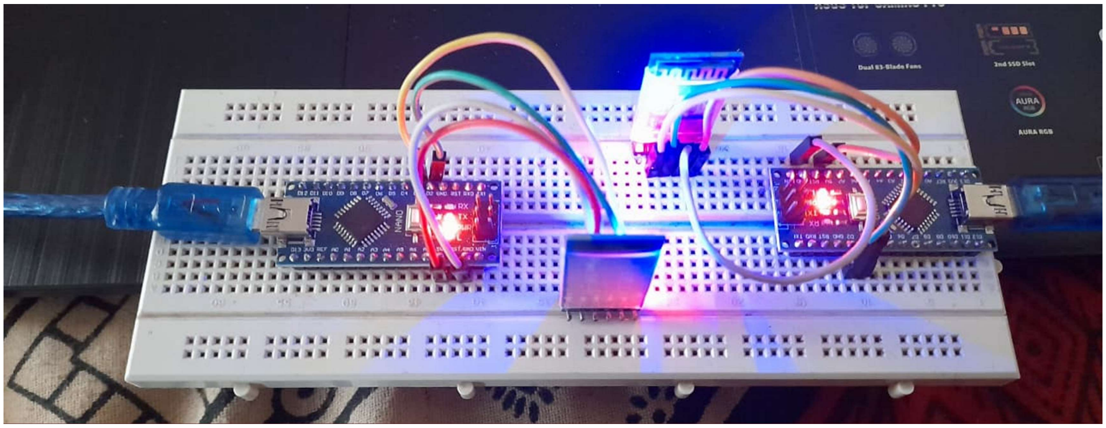

Figure 3.3: Publisher-subscriber communication using Arduino Nano and HC-05

The fourth stage involved implementing the fire detection system. A hybrid fire detection method was adopted instead of using only position-based detection. The fire object was provided with a detection zone using a sphere collider. When the drone enters this zone, the system recognizes that the drone is close enough to the possible fire location. A separate fire raycast target was then used to confirm whether the drone has a valid line of sight to the fire. Fire is confirmed only when the drone is inside the detection zone and the raycast successfully reaches the fire target. This method improves reliability because the drone does not confirm fire simply because it reached a coordinate. It must also verify the fire through proximity and raycast confirmation.

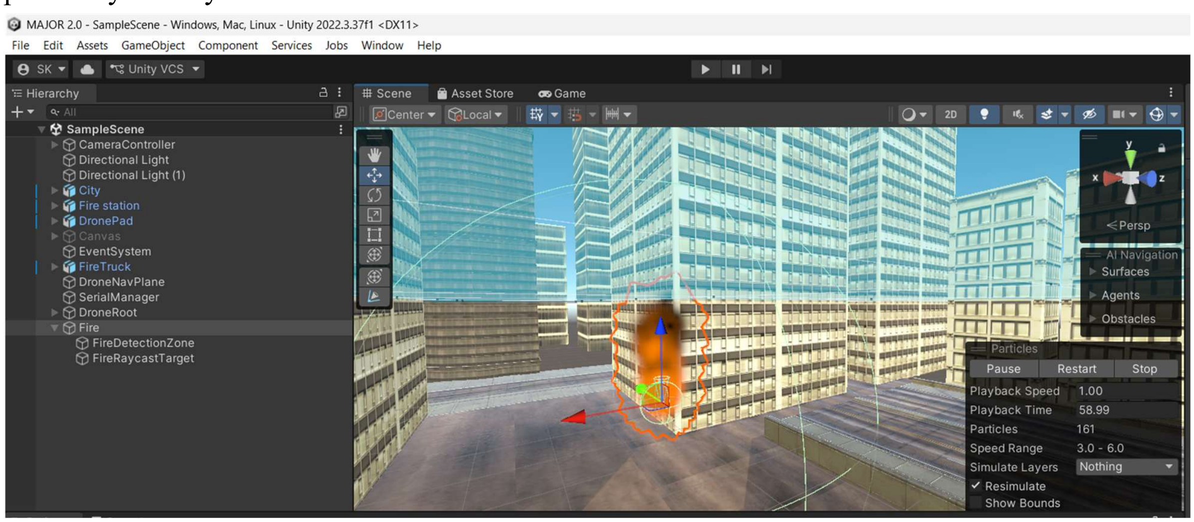

Figure 3.4: Hybrid fire detection using trigger zone and raycast target

The fifth stage involved fire truck deployment and drone landing. Once fire is confirmed, Unity deploys the fire truck near the fire location. The position of the truck is adjusted using an offset value so that it appears near the fire but not exactly inside the fire object. A landing point was added as a child object of the fire truck. This landing point acts as the final landing position for the drone. After the truck is deployed, the drone moves toward the truck and lands on the defined landing point. This completes the simulated emergency response workflow. After landing, Unity sends the mission complete status to MATLAB.

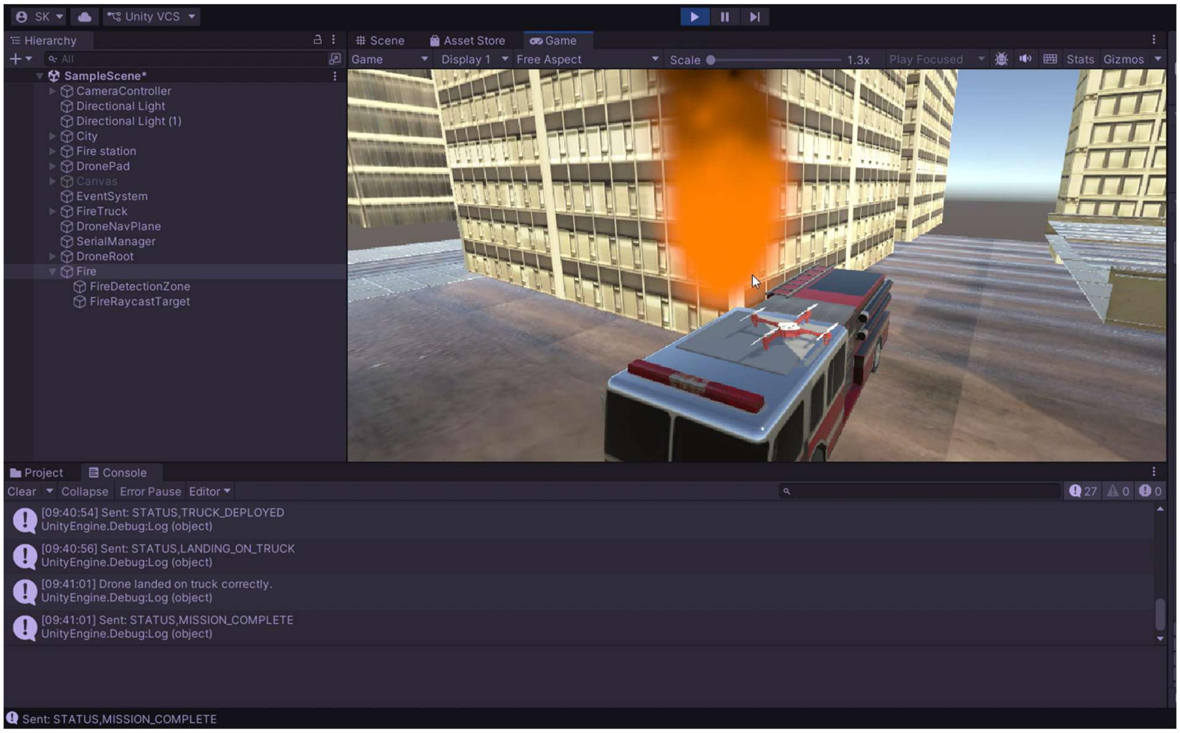

Figure 3.5: Fire truck deployment and drone landing point arrangement

The final stage was testing and debugging. Several issues were identified and corrected during testing. These included serial port access errors, coordinate parsing issues, drone teleportation, NavMesh placement errors, drone passing through buildings, fire detection failure, raycast obstruction, truck floating, and drone landing misalignment. Each issue was solved by adjusting the script logic, object hierarchy, collider setup, NavMesh settings, layer configuration, and object positions. The final system was tested until the drone successfully received the coordinate, navigated toward the fire, avoided buildings, detected the fire, deployed the truck, landed correctly, and returned a mission completion message.

3.2 Assumptions Made

Certain assumptions were made during the development of the project to simplify the design and ensure that the system could be demonstrated within a simulation environment. The fire location was assumed to be represented by a coordinate sent from MATLAB. In the present system, the fire coordinate is predefined or manually selected rather than detected from a real sensor or camera feed. This assumption was made because the primary aim of the project is to demonstrate the communication, navigation, verification, and response workflow.

It was also assumed that the city environment in Unity represents a simplified urban region. The buildings and roads are used mainly for navigation and obstacle avoidance demonstration. The drone is assumed to fly at a fixed height above the city, and the NavMesh navigation plane is used as the flight layer. Real drone aerodynamics, wind effects, battery constraints, camera noise, and GPS errors are not considered in this version of the project.

These factors can be included in future improvements if the system is extended to real UAV implementation.

Another assumption is that the HC-05 Bluetooth communication is suitable for short-range communication between the hardware circuit and the computer system. Since the project is a laboratory-level prototype, Bluetooth communication is sufficient for demonstrating the publisher-subscriber model. For real-world implementation, long-range wireless communication such as Wi-Fi, LoRa, GSM, or IoT-based cloud communication would be more suitable.

The fire detection zone and raycast target are assumed to represent the sensing region and visual confirmation point of a real drone-mounted fire detection system. In a real application, this may be replaced by thermal cameras, smoke sensors, flame sensors, or AI-based image processing. However, for simulation purposes, the trigger and raycast method provides a reliable and understandable way to represent fire verification.

3.3 Module Specifications

The project is divided into five major modules: communication module, Unity simulation module, drone navigation module, fire detection module, and response deployment module. The communication module is responsible for transferring coordinates and mission status messages between MATLAB and Unity. MATLAB sends the fire coordinate using serial communication, and Unity receives and parses the message. Unity then sends back mission updates so that the mission progress can be monitored from MATLAB.

The Unity simulation module consists of the city model, drone pad, drone, fire object, fire truck, roads, buildings, and navigation plane. This module provides the visual and functional environment for the project. The drone navigation module uses Unity NavMeshAgent for autonomous movement. The drone moves from the drone pad to the fire location, avoids buildings, hovers near the fire, and moves toward the truck landing point after fire confirmation.

The fire detection module consists of a fire detection zone and a fire raycast target. The detection zone is implemented using a trigger collider, while the raycast target is used to confirm line-of-sight detection. The fire is confirmed only if both conditions are satisfied. The response deployment module is responsible for placing the fire truck near the fire location and guiding the drone to land on the truck's landing point.

The modules are designed to work together in a sequential workflow. The communication module initiates the mission, the simulation module displays the environment, the navigation module moves the drone, the detection module verifies the fire, and the response module deploys the truck. This modular structure makes the system easier to debug, modify, and expand.

3.4 Type and Nature of Project

The project is a mixed-domain project involving software development, hardware communication, simulation, and experimental testing. The major part of the project is software-based because Unity is used to create the three-dimensional environment, implement drone navigation, perform fire detection, and simulate the response mechanism. The project also involves MATLAB programming for sending coordinates and receiving mission status messages.

The project includes hardware implementation through the use of Arduino Nano and HC-05 Bluetooth modules. The hardware part is mainly used to demonstrate communication between systems using a publisher-subscriber model. The circuit acts as a communication layer that supports the transfer of data between MATLAB and Unity. Therefore, the project is not purely a simulation project; it also includes embedded communication and hardware-level experimentation.

The nature of the project can be described as a simulation-based prototype of a cyber-physical emergency response system. It does not involve the fabrication of a real drone or fire truck, but it demonstrates the working logic of a drone-based fire detection and response workflow. The project includes software development, communication testing, navigation testing, and integrated system validation. This makes it suitable as an academic prototype for smart city emergency response and UAV-based disaster management applications.

3.5 Hardware and Software Requirements

The hardware requirements of the project include Arduino Nano boards, HC-05 Bluetooth modules, jumper wires, USB cables, and a computer system capable of running MATLAB and Unity. The Arduino Nano is used because it is compact, low-cost, and suitable for serial communication experiments. The HC-05 Bluetooth module is used because it supports wireless serial communication and can be configured for communication between embedded systems. Jumper wires and USB cables are used for circuit connections and programming.

The software requirements include Unity 2022.3.37f1, MATLAB, Arduino IDE, and the required Unity navigation components. Unity is used to create the 3D simulation environment and implement autonomous drone movement. MATLAB is used for transmitting fire coordinates and receiving status updates. Arduino IDE is used for programming and testing the Arduino Nano and HC-05 communication system. Unity's NavMesh system is used to implement path planning and obstacle avoidance.

The important Unity components used in this project include NavMeshSurface, NavMeshAgent, colliders, trigger zones, raycasting, transform control, and C# scripts. The NavMeshSurface is used to create the drone's flight navigation layer. The NavMeshAgent is used to move the drone automatically toward the target location. Colliders are used for physical boundaries and trigger detection. Raycasting is used for line-of-sight fire confirmation. C# scripts are used to control mission sequence, serial communication, drone movement, fire detection, truck deployment, and landing.

The selection of Unity is justified because it provides real-time 3D simulation, object control, physics support, NavMesh navigation, and visual debugging tools. It allows the system to be tested repeatedly without risk, cost, or damage. MATLAB is selected because it provides a reliable environment for numerical computation, serial communication, and data monitoring.

Arduino Nano is selected because it is simple, compact, and easy to integrate with serial communication modules. HC-05 is selected because it provides a low-cost wireless communication method suitable for academic demonstration.

The method and tool selection are appropriate for the project because the aim is to demonstrate a complete working model rather than build a full-scale real emergency response system. Unity provides the simulation environment, MATLAB provides the control and monitoring interface, Arduino Nano and HC-05 provide the embedded communication layer, and C# scripts connect all mission functions. This combination allows the system to demonstrate software integration, hardware communication, autonomous navigation, fire verification, and response deployment in a single project.

Table 3.1: Tools and Components Used

Tool / Component	Purpose in the project
Unity 2022	Creation of city scene, drone mission control, NavMesh navigation, raycast detection and visual response
C# scripts	Serial communication, mission controller logic, fire sensor logic and camera-follow behavior
MATLAB	Sending FIRE coordinate, receiving Unity status messages and displaying final mission state
Arduino Nano	Embedded board used for publisher-subscriber communication circuit
HC-05 Bluetooth module	Wireless serial communication between embedded units
NavMeshAgent	Autonomous movement of drone root over the elevated navigation plane
Colliders and triggers	Fire proximity detection, raycast targets and obstacle interaction
Fire truck and landing point	Response vehicle deployment and drone landing demonstration

CHAPTER 4
CONTRIBUTION OF EACH STUDENT

This chapter presents the contribution of each student involved in the project work. The project was carried out as a collaborative group activity, but each student was responsible for specific modules based on the requirements of the system. The complete project required the integration of Unity simulation, drone navigation, MATLAB communication, Arduino Nano and HC-05 based wireless communication, testing, debugging, result analysis, and report preparation. Therefore, the work was divided in such a way that both students could contribute individually as well as collectively toward the successful completion of the project.

The project team consisted of two students: **Spandan Kumar** and **H Anupam Nair**. Spandan Kumar mainly contributed to the Unity simulation side of the project, including the development of the city environment, drone object structure, NavMesh navigation, drone movement logic, obstacle avoidance, fire detection integration, truck deployment setup, and final simulation testing. H Anupam Nair mainly contributed to the MATLAB communication side of the project, including serial communication setup, fire coordinate publishing, Unity status receiving loop, and mission report display. Both students jointly worked on the Arduino Nano and HC-05 Bluetooth publisher-subscriber circuit, pairing, serial testing, communication validation, documentation, literature study, testing support, result analysis, and report preparation.

4.1 Contribution of Spandan Kumar

Spandan Kumar contributed mainly to the Unity-based simulation and autonomous drone mission implementation of the project. His work focused on developing the visual and functional environment required to demonstrate a drone-based fire detection and response system. The Unity part of the project was a major component because it visually represented the city, drone, fire object, fire truck, drone pad, obstacles, and the complete response sequence. The simulation environment had to be designed in such a way that the movement of the drone, detection of fire, and deployment of the truck could be clearly observed and tested.

One of the major contributions was the development of the Unity city environment. The city environment included roads, buildings, fire station, drone pad, fire object, fire truck, and other urban elements. The purpose of creating this environment was not only to make the project visually clear but also to provide a proper test area for drone navigation and obstacle avoidance. Buildings and roads were placed in the scene to simulate an urban layout. This helped in demonstrating how a drone could navigate through a complex city environment and move toward a fire location without direct manual control.

Spandan also worked on organizing the drone object structure in Unity. During the development process, it was observed that directly applying movement logic to the drone model could lead to issues such as incorrect rotation, unstable movement, unwanted offsets, and visual misalignment. To solve this, a separate root object called DroneRoot was created.

The main movement script and navigation components were attached to the root object, while the visible drone model was placed as a child object. This structure made the drone movement more stable and helped separate the functional movement logic from the visual model. This was an important design decision because it improved the reliability of the simulation.

Another important contribution was the implementation of NavMesh-based drone navigation. A drone navigation plane was created above the city at a fixed height. This plane acted as the flight layer for the drone. Unity's NavMesh system was used to allow the drone to move from the drone pad to the fire location while avoiding buildings and other obstacles. The NavMeshAgent component was configured on the drone root object to control the drone's pathfinding behavior. This allowed the drone to take off, reach the flight layer, move toward the fire coordinate, avoid obstacles, and later move toward the truck landing point.

Spandan also worked on the drone movement sequence. The drone was programmed to start from the drone pad, take off to a fixed height, move toward the received fire location, hover near the fire, verify the fire, and then proceed according to the mission result. If the fire was confirmed, the system deployed the fire truck and guided the drone to land on the truck. If fire was not confirmed, the drone returned to its starting point. This movement sequence was tested and refined several times to ensure that the drone did not teleport, pass through buildings, move below the city, or become misaligned with the navigation plane.

The fire detection integration inside Unity was also supported through the simulation setup. The fire object was provided with a detection zone and raycast target so that the drone could verify the fire using a hybrid method. The trigger zone was used to detect whether the drone had entered the fire region, while the raycast target was used to confirm line-of-sight detection. The system was adjusted until the drone could correctly detect the fire and send the appropriate mission status. This process involved testing different positions, layers, colliders, and raycast configurations.

Spandan also contributed to the fire truck deployment and landing arrangement in Unity. After fire confirmation, the fire truck had to appear near the fire location and remain properly placed on the city ground. A landing point was added to the truck so that the drone could land on the top of the vehicle instead of entering the truck or landing at an incorrect position. This part required several adjustments to the truck offset, ground placement logic, and landing point arrangement. The final setup allowed the truck to be deployed near the fire and the drone to land on it successfully.

Final simulation testing was another major contribution. The Unity system was tested repeatedly to check whether the complete mission workflow was functioning correctly. This included checking whether Unity received the fire coordinate, whether the fire object was placed correctly, whether the drone started its mission, whether the drone avoided obstacles, whether the fire detection worked, whether the truck deployed properly, whether the drone landed correctly, and whether the mission completed successfully. Many issues were identified during testing, such as serial port errors, coordinate parsing errors, drone teleportation, NavMesh placement problems, fire detection failure, raycast obstruction, truck floating, and landing misalignment. These issues were solved step by step through debugging and refinement.

Overall, Spandan Kumar’s contribution was mainly related to the Unity simulation, drone navigation, mission execution, and final working demonstration of the project. His work helped convert the project concept into a visible and functional simulation. The Unity implementation formed the main output of the project because it demonstrated the complete drone-based fire detection and response workflow in a realistic 3D environment.

4.2 Contribution of H Anupam Nair

H Anupam Nair contributed mainly to the MATLAB-based communication and mission monitoring part of the project. His work focused on creating the control-side program that sends the fire coordinate to Unity and receives the mission status messages from Unity. MATLAB played an important role in the project because it acted as the source of the fire coordinate and the monitoring interface for the complete mission. Without this communication module, Unity would not receive the required fire location data and the mission would not begin automatically.

One of the major contributions of H Anupam Nair was the implementation of MATLAB serial communication. The serial communication setup was necessary for transferring data between MATLAB and the external communication system. The MATLAB script was written to open the required COM port, configure the baud rate, set the terminator, flush the serial buffer, and prepare the system for data transmission. This setup ensured that MATLAB could send structured messages and receive responses from the Unity side through serial communication.

Another important contribution was the development of the FIRE coordinate publishing system. The MATLAB program was designed to publish the fire location using a structured message format. The fire coordinate was represented using X, Y, and Z position values. These values were converted into a message such as FIRE,x,y,z and transmitted through the serial port. This allowed the Unity side to identify the message type, extract the coordinate values, and begin the drone mission. The coordinate publishing system was important because it made the project workflow data-driven instead of manually controlled inside Unity.

H Anupam also worked on the status receiving loop in MATLAB. After Unity receives the fire coordinate and starts the mission, it sends back different mission status messages. The MATLAB script was designed to continuously check whether new data was available on the serial port. When data was received, the script read the message, cleaned the text, split it into parts, and identified the message type. This allowed MATLAB to understand different mission states such as acknowledgement, movement status, fire confirmation, truck deployment, returning home, landing, and mission completion.

The mission report display was also implemented on the MATLAB side. The program-maintained mission state variables such as acknowledgement received, fire reached, fire confirmed, returned home, and mission complete. At the end of the mission, MATLAB displayed a final mission report showing the result of the simulation. This made it possible to verify whether the mission was completed successfully from the control side. The report display also helped during debugging because it showed which stage of the mission had been completed and where any failure might have occurred.

H Anupam's work was important for establishing the link between the external control system and the Unity simulation. The MATLAB script acted as the command and monitoring interface. It allowed the project to demonstrate that the drone mission could be initiated from outside Unity using serial communication. This made the project more realistic because in real-world applications, drone missions are often triggered by data from external sensors, control stations, or communication networks.

The MATLAB communication module also helped in validating two-way communication. It was not enough for MATLAB to only send the coordinate; Unity also had to respond with mission updates. H Anupam's receiving loop made this possible by continuously reading incoming messages from Unity. This created a feedback mechanism where the control side could monitor the mission progress. The feedback messages improved the reliability of the system because the user could know whether the drone had received the coordinate, whether fire was confirmed, whether the truck was deployed, and whether the mission was completed.

During testing, H Anupam contributed to identifying communication-related problems. These included COM port selection, baud rate configuration, message terminator settings, serial buffer issues, and message formatting. Serial communication can fail if the port is already in use, if the baud rate does not match, or if the message terminator is not properly configured. These issues were checked and corrected during the testing phase. The final communication setup allowed MATLAB and Unity to exchange messages reliably.

H Anupam also contributed to making the MATLAB code readable and structured. The code was arranged into sections for serial settings, fire location definition, mission state variables, main loop, message parsing, status handling, and final report. This structure made it easier to understand the operation of the code and modify it when required. It also made the code suitable for demonstration and documentation in the project report.

Overall, H Anupam Nair's contribution was mainly related to MATLAB serial communication, fire coordinate publishing, mission status receiving, and mission report generation. His work provided the control and monitoring side of the project. The MATLAB implementation allowed the Unity drone simulation to be triggered externally and monitored throughout the mission. This contribution was essential for completing the communication and control aspect of the drone-based fire detection and response system.

4.3 Group Contribution

In addition to individual responsibilities, both students contributed jointly to several important parts of the project. The project required close coordination between the Unity simulation side and the MATLAB communication side. Therefore, both students worked together to make sure that the fire coordinate sent from MATLAB was correctly received by Unity and that Unity's mission status messages were correctly displayed in MATLAB. This integration required repeated testing, debugging, and communication validation.

A major group contribution was the development and testing of the Arduino Nano and HC-05 Bluetooth publisher-subscriber communication circuit. Both students worked on connecting the Arduino Nano with the HC-05 Bluetooth module and testing serial communication. The purpose of this circuit was to demonstrate an embedded wireless communication layer in the project. The publisher-subscriber model was used to represent communication between different system components. MATLAB acted as the publisher of the fire coordinate, while Unity acted as the subscriber that received and used the coordinate to begin the mission. Unity then published mission status messages, which MATLAB received and displayed.

Both students contributed to HC-05 Bluetooth pairing and communication validation. This involved checking whether the Bluetooth modules were powered correctly, paired properly, and communicating through the correct serial port. Serial testing was performed to verify whether messages were being transmitted and received without error. The students also checked the COM port configuration, baud rate, and message format. This was an important part of the project because communication failure would prevent the complete system from working as expected.

Documentation and literature study were also carried out as group work. Both students referred to existing work related to drone-based fire detection, UAV applications in disaster management, wireless communication, simulation-based testing, and autonomous navigation. The literature study helped in understanding the present state of the work area and identifying the need for an integrated simulation-based prototype. Both students contributed to organizing the collected information and relating it to the project objectives.

Testing support was another important group contribution. The complete system required several rounds of testing because the project involved multiple modules. MATLAB communication, Unity coordinate parsing, drone movement, NavMesh navigation, obstacle avoidance, fire detection, truck deployment, and drone landing had to work together. When one part failed, the entire mission sequence was affected. Therefore, both students supported the testing process by observing outputs, identifying errors, discussing possible causes, and verifying corrections.

Both students also contributed to result analysis. After the system started working correctly, the results were analyzed based on whether the project objectives were achieved. The analysis included checking whether the coordinate was transmitted successfully, whether the drone reached the fire area, whether the detection system confirmed the fire, whether the truck was deployed correctly, and whether the mission complete status was received. The final result showed that the proposed system was able to demonstrate the intended drone-based fire detection and response workflow.

Report preparation was also completed as a group activity. Both students contributed to organizing the project content, explaining the methodology, listing hardware and software requirements, preparing result discussions, and including suitable figures and screenshots. The report was prepared to present the project in a formal academic format. The contribution of both students helped ensure that both the technical implementation and documentation aspects of the project were completed properly.

Overall, the group contribution was essential for the successful completion of the project. While individual modules were handled separately, the final project required integration and teamwork. The combined work of both students resulted in a complete system that includes MATLAB communication, Arduino Nano and HC-05 publisher-subscriber communication, Unity simulation, autonomous drone navigation, hybrid fire detection, fire truck deployment, drone landing, and mission status reporting.

CHAPTER 5
RESULTS AND DISCUSSION

This chapter presents the results obtained from the drone-based fire detection and response system. The results are analyzed based on the successful working of the major modules of the project, including MATLAB communication, Arduino Nano and HC-05 publisher-subscriber communication, Unity simulation, NavMesh-based drone navigation, obstacle avoidance, hybrid fire detection, fire truck deployment, drone landing, and mission status reporting. The system was tested as a complete workflow, starting from fire coordinate transmission and ending with mission completion.

The main purpose of the result analysis is to verify whether the project objectives were achieved. The expected result was that MATLAB should send the fire coordinate, Unity should receive the coordinate, the drone should move toward the fire location, the drone should avoid buildings, the fire should be detected using trigger and raycast confirmation, the fire truck should be deployed near the fire, the drone should land on the truck, and MATLAB should receive the final mission status. After repeated testing and debugging, the final system performed the required sequence successfully.

5.1 Result Analysis

The working of the project was analyzed by dividing the system into different functional stages. Each stage was tested separately and then verified as part of the complete integrated system. The major stages tested were coordinate transmission, message reception in Unity, drone take-off, autonomous navigation, obstacle avoidance, fire detection, truck deployment, drone landing, and mission completion.

The following table shows the result analysis of each major module.

Table 5.1: Functional result analysis of the project system

Sl. No.	Module / Function Tested	Expected Result	Observed Result	Status
1	MATLAB Fire coordinate publishing	MATLAB should send FIRE coordinate through serial communication	FIRE coordinate was successfully sent from MATLAB	Successful
2	Unity serial message reception	Unity should receive and parse the coordinate	Unity received the coordinate and started the mission	Successful
3	Arduino Nano and HC-05 communication	Wireless publisher-subscriber communication should support serial data transfer	HC-05 pairing and serial testing were completed	Successful

Sl. No.	Module / Function Tested	Expected Result	Observed Result	Status
4	Drone take-off	Drone should rise from the drone pad to fixed flight height	Drone successfully moved to the required flight height	Successful
5	NavMesh-based movement	Drone should move toward the fire location using NavMeshAgent	Drone moved toward the received coordinate	Successful
6	Obstacle avoidance	Drone should avoid buildings instead of passing through them	Drone avoided buildings after NavMesh obstacle setup	Successful
7	Fire trigger detection	Drone should identify when it enters fire detection zone	Trigger zone detected drone presence near fire	Successful
8	Raycast fire confirmation	Raycast should confirm the fire target	Fire was confirmed after correcting raycast target and layer settings	Successful
9	Fire truck deployment	Truck should appear near fire location and remain on ground	Truck was deployed near fire after adjusting offset and placement logic	Successful
10	Drone landing	Drone should land on truck landing point	Drone landed correctly on the truck after landing point adjustment	Successful
11	Mission status reporting	Unity should send mission status to MATLAB	MATLAB received status messages such as fire confirmed, truck deployed, and mission complete	Successful
12	Final mission completion	Complete workflow should end without manual intervention	Full mission sequence worked as intended	Successful

From Table 5.1, it can be observed that all major project modules were completed successfully after testing and correction. The initial testing produced several errors, but these were solved through step-by-step debugging. The final integrated system was able to perform the complete mission sequence as planned.

The communication result shows that MATLAB was able to publish the fire coordinate in a structured format. Unity received the coordinate and used it to start the drone mission. The use of a structured message such as FIRE,x,y,z made it possible for Unity to identify the type of data received and extract the coordinate values correctly. The status messages sent from Unity to MATLAB also helped in monitoring the progress of the mission.

The navigation result shows that the drone was able to fly toward the fire location using Unity's NavMeshAgent system. In the initial stages, the drone passed through buildings or moved incorrectly because of NavMesh and object hierarchy issues. These problems were solved by creating a proper DroneRoot structure, using a drone navigation plane, configuring NavMesh surfaces, and adjusting obstacle settings. After these corrections, the drone moved smoothly toward the fire while avoiding buildings.

The fire detection result shows that the hybrid trigger and raycast method worked successfully. The trigger zone alone was not considered sufficient because it could confirm fire only based on proximity. Therefore, a raycast target was used to confirm that the drone had a valid line of sight toward the fire. The final result showed that fire was confirmed only when both detection conditions were satisfied. This improved the reliability of the simulated detection process.

The response result shows that the fire truck was deployed after fire confirmation. During testing, the truck initially floated or appeared at an incorrect height. This issue was corrected by adjusting the truck offset and placement logic. A landing point was also added to the truck so that the drone could land properly on the vehicle instead of going inside it. The final simulation showed that the truck deployment and drone landing sequence worked as required.

5.2 Graphical and Tabular Result Representation

The result of the complete mission can be represented as a sequence of mission states. Each state indicates one important step in the project workflow. The following table shows the mission sequence observed during the final working test.

Table 5.2: Mission sequence and status output

Mission Stage	**System Action**	**Output / Status Message**
Stage 1	MATLAB sends fire coordinate	FIRE,x,y,z
Stage 2	Unity receives coordinate	Coordinate parsed successfully
Stage 3	Drone starts mission	Drone take-off initiated
Stage 4	Drone moves to fire location	Moving toward target
Stage 5	Drone enters fire detection zone	Trigger detected
Stage 6	Raycast checks fire target	Fire target confirmed
Stage 7	Fire is verified	STATUS,FIRE_CONFIRMED
Stage 8	Fire truck is deployed	STATUS,TRUCK_DEPLOYED
Stage 9	Drone moves to truck landing point	Landing sequence started
Stage 10	Drone lands on truck	Drone landed correctly
Stage 11	Mission ends	STATUS,MISSION_COMPLETE

Table 5.2 shows that the project follows a clear sequence of operations. Each stage depends on the successful completion of the previous stage. For example, the drone mission cannot begin until Unity receives the coordinate, and the truck cannot be deployed until fire is confirmed. This shows that the project works as an integrated system rather than as separate disconnected modules.

The result can also be represented in graphical form using a simple success analysis of the main project objectives.

Table 5.3: Objective-wise result summary

Project Objective	**Result Achieved**	**Remarks**
Development of Unity city environment	Achieved	City, fire station, drone pad, fire, truck, and buildings were included
MATLAB to Unity coordinate communication	Achieved	Fire coordinate was sent and received successfully
Arduino Nano and HC-05 communication model	Achieved	Publisher-subscriber communication was tested
Autonomous drone navigation	Achieved	Drone moved automatically using NavMeshAgent
Obstacle avoidance	Achieved	Drone avoided buildings after NavMesh correction
Hybrid fire detection	Achieved	Fire was confirmed using trigger zone and raycast
Fire truck deployment	Achieved	Truck was deployed near fire location
Drone landing on truck	Achieved	Drone landed on the truck landing point
Mission status feedback	Achieved	MATLAB displayed mission status messages

The tabular analysis shows that all main objectives of the project were achieved. The project successfully demonstrates a simulation-based prototype of a drone-assisted fire detection and response system. The results also show that Unity, MATLAB, Arduino Nano, and HC-05 can be combined to represent a cyber-physical workflow.

5.3 Explanation of the Results

The final result of the project confirms that the proposed system can complete a fire response mission in a simulated city environment. MATLAB acts as the control and monitoring side by sending the fire coordinate and receiving mission updates. Unity acts as the simulation and mission execution side by controlling the drone, fire, truck, navigation, and detection. The Arduino Nano and HC-05 modules demonstrate the hardware communication layer used for publisher-subscriber communication.

The drone movement result is significant because the drone is not manually controlled after receiving the coordinate. Once the mission starts, the drone automatically takes off, moves toward the target, avoids buildings, checks the fire, and completes the response sequence. This shows that the project satisfies the autonomous workflow requirement. Although it is a simulation, the logic represents how an actual drone-based emergency response system may work at a prototype level.

The fire detection result is also important because the project does not use only one detection condition. A single condition such as distance may produce false results. For example, the drone may be near the fire coordinate but may not have a clear view of the fire. To reduce this issue, the project uses a hybrid method. The trigger zone checks proximity, while the raycast target checks line-of-sight confirmation. Fire is confirmed only when both are satisfied. This makes the fire verification process more reliable than simple distance-based detection.

The fire truck deployment and landing result shows that the response phase is also included in the project. Many fire detection simulations end after detecting the fire, but this project continues the workflow by deploying a response vehicle and landing the drone. This makes the project more complete because it includes detection, confirmation, response, and mission completion.

5.4 Significance of the Result Obtained

The result obtained is significant because it demonstrates a complete end-to-end fire detection and response workflow. The system does not only show drone movement or only show communication. It combines multiple areas such as simulation, wireless communication, navigation, obstacle avoidance, detection, and emergency response. This gives the project a practical structure and makes it relevant to smart city emergency management concepts.

The result also shows the importance of simulation in developing and testing emergency response systems. Testing a real drone in a real fire environment is risky, expensive, and difficult. Unity simulation allows the same logic to be tested safely in a virtual environment. The drone can be moved, obstacles can be added, errors can be observed, and the mission can be repeated multiple times without physical damage or safety risks.

Another significant result is the integration between MATLAB and Unity. MATLAB is commonly used for computation, data handling, and control logic, while Unity is commonly used for simulation and visualization. By connecting these two tools through serial communication, the project demonstrates how engineering computation and real-time simulation can be combined. The Arduino Nano and HC-05 communication circuit further adds an embedded systems component to the project.

The hybrid detection result is significant because it improves reliability. In emergency response systems, false detection and missed detection are both serious problems. The use of trigger and raycast together provides a simple but effective method for improving confirmation accuracy in simulation. This concept can later be extended using real sensors such as flame sensors, thermal cameras, smoke sensors, or AI-based vision systems.

5.5 Deviations from Expected Results and Justification

During the development and testing of the project, some deviations were observed from the expected results. These deviations were mainly due to integration issues between Unity objects, NavMesh settings, serial communication configuration, collider behavior, and object positioning. However, these issues were identified and corrected during debugging.

One of the initial deviations was that the drone teleported or disappeared after receiving coordinates. This occurred because of incorrect object hierarchy and movement handling. The issue was solved by using a separate DroneRoot object and placing the visual drone model as its child. This allowed movement scripts and NavMeshAgent to control the root object while the visual drone remained properly aligned.

Another deviation was that the drone initially passed through buildings instead of avoiding them. This happened because the NavMesh surface and obstacle configuration were not properly set. The issue was solved by adjusting the drone navigation plane, baking the NavMesh correctly, and configuring obstacles so that the drone followed a valid path instead of passing through buildings.

A further deviation was that the fire was not detected even when the drone reached the fire area. This was due to problems related to the trigger zone, raycast target, layer settings, and raycast obstruction. The detection method was corrected by using a fire detection zone for proximity and a separate fire raycast target for line-of-sight confirmation. After adjusting the layers and target object, the drone successfully detected the fire.

The fire truck also showed positioning issues during testing. At first, the truck appeared floating or at an incorrect position. This was caused by the difference between the fire coordinate, city ground height, model pivot point, and truck offset values. The problem was reduced by adjusting the truck deployment logic and offset values. The landing issue was solved by adding a separate landing point on the truck so that the drone landed on the correct position.

These deviations were expected because the project involved integration between multiple systems and tools. Unity object hierarchy, NavMesh movement, serial communication, raycasting, and collider-based detection all had to work together. The debugging process helped improve the system and made the final implementation more stable. The final result satisfies the intended project objectives.

5.6 Group Analysis

The project result shows that the work was successfully completed through the combined contribution of both students. The Unity side, MATLAB side, and hardware communication side had to be tested together to make the final system work correctly. The group analysis shows that the project required both individual module development and collaborative integration.

Unity simulation work provided the visual and functional environment for the project. The drone navigation, fire detection setup, truck deployment, and final simulation testing were essential for demonstrating the project output. MATLAB communication work provided the control and monitoring side of the project. The coordinate publishing, status receiving loop, and mission report display made the system interactive and data-driven. Both students contributed to the Arduino Nano and HC-05 communication circuit, serial testing, documentation, result analysis, and report preparation.

The final result confirms that the division of work was effective. Each student handled a major part of the system, and the combined effort resulted in a complete working prototype. The project successfully demonstrates the concept of using a drone for fire detection and response in a simulated smart city environment.

CHAPTER 6
CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The project successfully developed a drone-based fire detection and response simulation using MATLAB, Arduino Nano, HC-05 Bluetooth communication and Unity. The system demonstrated the full workflow from coordinate publishing to drone movement, fire verification, truck deployment and mission completion. The use of MATLAB provided a control and monitoring interface, while the Arduino Nano and HC-05 modules represented the embedded communication layer. Unity provided the visual and operational environment required to demonstrate the autonomous response.

The final Unity implementation solved several important system integration problems. The drone moved correctly after the DroneRoot and visual child structure was established. The NavMeshAgent enabled autonomous navigation and obstacle avoidance at the selected flight height. The hybrid trigger and raycast detection method successfully confirmed the fire. The response vehicle was deployed near the fire and the drone landing point allowed the drone to land correctly on the truck. These results show that the project objectives were achieved.

The project also provided practical learning in serial communication, Unity scripting, object hierarchy, NavMesh configuration, collider and trigger setup, raycast detection, simulation debugging and team-based integration. The work is suitable as a foundation for future improvement because each subsystem can be upgraded independently without changing the overall mission concept.

6.2 Future Scope

The project can be extended by replacing the simulated fire confirmation with camera-based image processing. A virtual camera in Unity or a real camera on a drone can be used to detect flame and smoke using computer vision or deep learning. This would make the detection subsystem closer to practical UAV fire monitoring systems.

Another future extension is the integration of real UAV hardware. The current Unity drone can be connected to a physical drone control system using telemetry data, GPS coordinates and wireless communication. The simulation can then act as a digital twin or test environment for mission planning before real flight.

The communication system can also be improved. HC-05 Bluetooth is useful for demonstration but has limited range and bandwidth. Future versions can use Wi-Fi, LoRa, Zigbee, cellular communication or IoT cloud platforms for longer range and more reliable emergency communication. The MATLAB side can be expanded into a full monitoring dashboard with live maps, logs and mission analytics.

The fire truck response can be upgraded into a more detailed emergency response model. Multiple response vehicles, road-based navigation, water-spraying animation and traffic constraints can be added. A dispatch algorithm can be created to choose the nearest available truck based on distance and accessibility.

The city environment can be expanded into a smart-city simulation. Multiple fire sensors, CCTV inputs, building alarms, drone stations and control centers can be included. The system can also support multiple drones for cooperative search, verification and tracking. Such improvements would make the project more useful for research in urban emergency automation and disaster management.

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Annexure 1

PO &PSO Mapping

Student Name: Spandan Kumar / H Anupam Nair
Registration No: 220929337 / 220929228

PO	✓ Tick	Page No	Section No	Supervisor Observation
PO1	✓	11–18, 26–33	3.1–3.5, 5.1–5.6	
PO2	✓	7–10, 11–14, 26–32	2.1–2.3, 3.1–3.2, 5.1–5.5	
PO3	✓	9–10, 12–18	2.3, 3.1–3.5	
PO4	✓	7–8, 26–33	2.1–2.2, 5.1–5.6	
PO5	✓	11–18	3.1–3.5	
PO6	✓	4–5	1.6–1.7	
PO7	✓	4–5, 14	1.6, 3.2	
PO8	✓	19–24	4.1–4.3	
PO9	✓	11–18, 26–33, 36–38	3.1–3.5, 5.1–5.6, References	
PO10	✓	4, 19–24	1.5, 4.1–4.3	
PO11	✓	7–8, 16–18, 34–35	2.1–2.2, 3.5, 6.2	

PSO	✓ Tick	Pg. No	Section No	Supervisors Observation
PSO1	✓	11–18	3.1–3.5	
PSO2	✓	12–18, 26–33	3.1–3.5, 5.1–5.6	
PSO3	✓	19–24, 34–35	4.1–4.3, 6.2	

Signature of Student: **Name and Signature of Supervisor:**

Spandan Kumar:

H Anupam Nair:

Date:

Annexure 2

LO Mapping

Student Name: Spandan Kumar / H Anupam Nair
Registration No: 220929337 / 220929228

LO	✓ Tick	Page No	Section No	Supervisor Observation
C1.	✓	11–18, 26–33	3.1–3.5, 5.1–5.6	
C2.	✓	7–10, 11–18, 26–32	2.1–2.3, 3.1–3.5, 5.1–5.5	
C3.	✓	11–18, 26–33	3.1–3.5, 5.1–5.6	
C4.	✓	7–10, 36–38	2.1–2.3, References	
C5.	✓	4–5, 11–18, 26–33	1.6–1.7, 3.1–3.5, 5.1–5.6	
C6.	✓	11–18, 26–33	3.1–3.5, 5.1–5.6	
C7.	✓	4–5, 34–35	1.6, 6.2	
C8.	✓	4–5, 14–18	1.6, 3.2–3.5	
C9.	✓	11–18, 26–33	3.1–3.5, 5.1–5.6	
C10.	✓	11–18, 26–33	3.1–3.5, 5.1–5.6	
C11.	✓	19–24	4.1–4.3	
C12.	✓	14–18, 26–33	3.3–3.5, 5.1–5.6	
C13.	✓	11–18	3.1–3.5	
C14.	✓	26–33, 34–35	5.1–5.6, 6.2	
C15.	✓	4–5, 19–24	1.5, 4.1–4.3	
C16.	✓	19–24	4.1–4.3	
C17.	✓	26–38	5.1–5.6, 6.1–6.2, References	
C18.	✓	7–18, 34–35	2.1–2.3, 3.1–3.5, 6.2	

Signature of Student: **Name and Signature of Supervisor:**

Spandan Kumar:

H Anupam Nair:

Date:

Annexure 3

Address of IET learning outcomes during the project period

1. Explain the steps you considered to investigate and define the problem in your project work *(C4, evaluate level)*

Answer: The problem was investigated by first identifying the limitations in conventional fire response workflow, where a fire is generally reported by a person or a fixed sensor before emergency action is taken. The major issue observed was the time delay between reporting, verification and response. Based on this, the project problem was defined as the need for a simulated autonomous system that can receive a fire coordinate, verify the fire through a drone, and then initiate response deployment. The team studied the project requirements, available tools, and feasibility of combining MATLAB, Arduino Nano, HC-05 Bluetooth communication and Unity simulation. The problem was then divided into smaller modules such as coordinate communication, drone navigation, fire detection, obstacle avoidance and truck deployment. This helped in clearly defining the scope and avoiding unnecessary complexity. The final problem statement was framed as the design and implementation of a drone-based fire detection and response simulation with embedded publisher-subscriber communication support.

2. Discuss the science, mathematics, statistics, engineering principles and other basic technology you identified for design (Mechanical, Electronic, Physics, Chemistry, Automation) in your project work. *(C1, C2, C3, Application, Analysis, Evaluation of Science and Mathematics in the project)*

Answer: The project required the application of several basic science, mathematics, engineering and technology principles. In mathematics, the system used three-dimensional coordinate geometry for representing fire position, drone position, navigation target and truck deployment location. Vector distance calculation was used to check whether the drone had reached the fire area or the landing location. In physics, the concept of motion, position, height, line-of-sight and raycasting was applied to simulate drone movement and fire verification. Mechanical design thinking was used while arranging the drone, truck, landing point and city environment so that movement and response actions appeared realistic. Electronics knowledge was used in the Arduino Nano and HC-05 Bluetooth publisher-subscriber circuit, including serial communication, pairing and data transfer. Automation principles were used because the drone carried out take-off, navigation, fire checking, truck deployment and landing without continuous manual control after receiving the coordinate. Software technology was applied through Unity, C# scripting, MATLAB serial programming and NavMesh-based path planning.

3. Have you considered the Environmental and Sustainability limitations in your project work? *(C7, evaluate)*

Answer: Yes, environmental and sustainability limitations were considered during the project. Since the work is a simulation-based prototype, it does not produce smoke, noise, fuel consumption or physical waste during testing. This makes the project safer and more sustainable for repeated academic demonstration compared with a real fire experiment. The concept itself also has environmental relevance because faster fire detection and response can reduce fire spread, smoke emission, damage to buildings and loss of resources. The use of simulation reduced the need for physical trials, thereby minimizing material usage and energy consumption. The hardware part was limited to low-power components such as Arduino Nano and HC-05 Bluetooth modules, which are suitable for laboratory-level communication testing. The sustainability limitation is that the present system does not yet include real UAV power optimization, battery management or environmental sensing. However, the project provides a foundation that can be extended toward efficient real-world emergency response systems in the future.

4. Have you considered ethical, health, safety, security, and risk issues; intellectual property; codes of practice and standards while addressing these issues in your project work? If so, Explain in detail. *(C5, create)*

Answer: Yes, ethical, health, safety, security, intellectual property, codes of practice and standards were considered while developing the project. Since the work involves a simulated emergency response system, false fire confirmation and unsafe response behavior were treated as important design concerns. To reduce false confirmation, the project used a hybrid fire detection approach involving both trigger-zone proximity and raycast-based line-of-sight confirmation. Hardware safety was considered by using low-voltage Arduino Nano and HC-05 modules, proper serial connection and careful testing. Health risks were minimized because no real flame or hazardous material was used. Security concerns were considered at a basic level because wireless communication should transmit structured messages and avoid random data interpretation. Intellectual property concerns were handled by using project-generated scripts, properly documenting external concepts and avoiding direct copying of proprietary work. The project followed general safe laboratory practice, responsible software testing, careful wiring and ethical reporting of results without exaggerating the system as a fully real-world fire-fighting solution.

5. What were the esthetical issues faced and how it was addressed in your project in the design phase? *(C5, analysis)*

Answer: The main aesthetical issues faced in the design phase were related to the visual arrangement of the Unity city, drone, fire object, fire truck and navigation plane. Initially, the drone model, DroneRoot object and drone pad alignment created visual offsets. In some tests, the drone appeared away from the pad, tilted incorrectly or moved in an unrealistic manner. These issues were addressed by separating the drone movement logic from the visual model through a parent-child structure. The DroneRoot controlled the mission logic, while the visual drone was adjusted as a child object for proper scale, rotation and placement. Another aesthetic issue was the appearance of the fire truck after deployment. The truck had to be positioned near the fire but not appear floating or underground. The landing point was added to make the drone landing visually clear. The fire object and detection zone were also arranged so that the fire remained visible and understandable during demonstration. These improvements made the simulation more presentable and easier to explain.

6. Were there any health issues considered during design process? How it was addressed in your project in the design phase? *(C5, create)*

Answer: Health issues were considered during the design process mainly from the perspective of safe laboratory implementation and emergency-response relevance. The project did not involve real fire, smoke, high-voltage equipment, rotating UAV propellers or heavy mechanical movement. Therefore, the immediate health risk to students during development was low. The hardware part used Arduino Nano and HC-05 Bluetooth modules, which operate at low voltage and are suitable for academic experimentation when handled properly. Care was taken to avoid incorrect wiring, overheating of components and unsafe handling of powered circuits. From the application point of view, the project supports human safety because a drone can verify a fire location before humans move into a potentially dangerous area. The simulated system demonstrates how remote verification can reduce exposure to smoke, heat and unsafe structures. The limitation is that the present model does not include medical or physiological sensors, but it still addresses health and safety indirectly by improving emergency awareness.

7. What were the safety, security and risk issues considered in the design stage? *(C10, create)*

Answer: The safety, security and risk issues considered in the design stage included communication failure, wrong coordinate interpretation, collision with buildings, false fire confirmation and unsafe truck deployment. To reduce navigation risk, the drone used NavMesh-based movement with obstacle avoidance so that it could travel around buildings instead of passing through them. To reduce detection risk, fire confirmation was not based only on reaching a coordinate; it also required trigger-zone detection and raycast verification. To reduce communication risk, the MATLAB-to-Unity message format was kept simple and structured using packets such as FIRE and STATUS. The Unity console was used for debugging received messages and mission states. Hardware risk was minimized by using low-voltage components and testing the HC-05 Bluetooth connection carefully. Security was considered by limiting the communication to the expected serial port and expected message format. Although the system is not designed for real deployment yet, these design decisions reduced risk in the simulated workflow.

8. Have you come across intellectual property issues in the project phase? *(C5, create)*

Answer: No major intellectual property issue was encountered during the project phase. The main project logic, Unity scene arrangement, C# scripts, MATLAB serial communication workflow and Arduino-HC-05 circuit testing were developed as part of the academic work. However, general concepts such as UAV-based monitoring, NavMesh navigation, serial communication and fire detection were studied from literature, documentation and available learning resources. These were used only for academic understanding and not copied as proprietary designs. Any images, models or assets used in the Unity environment must be credited properly if they are taken from external asset sources. The report references should also acknowledge literature and online resources used during development. The team avoided presenting externally available ideas as original inventions. At present, no patent or formal IPR filing has been completed. If the project is developed further with new algorithms, hardware integration or field implementation, possible intellectual property protection can be explored.

9. What are the codes of conduct and standards you needed to use in design phase and in other phases of your project as well? (It may include codes of practice and standards for safety, security, health, risk) Explain the legal issues, ISO standards, IEC standards, etc. *(C8, evaluate)*

Answer: The project followed general codes of practice related to safe laboratory work, software testing, documentation and responsible use of electronic components. In the hardware stage, the Arduino Nano and HC-05 modules were handled using low-voltage laboratory practices, proper wiring and safe serial testing. For communication, the project followed a structured serial message format so that transmitted and received data could be interpreted clearly. In the software stage, Unity and MATLAB were used according to standard development practices such as modular scripting, debugging, testing, and documenting the workflow. Since the system is a prototype simulation, formal aviation, fire safety and emergency-response standards were not directly implemented. However, the project concept is related to standards such as ISO 45001 for occupational health and safety management, ISO 14001 for environmental management, and IEC-related safe electrical practices. In a real implementation, UAV regulations, wireless communication rules, fire safety codes and data security requirements would need to be followed strictly.

10. What were the professional ethics needed to be followed in general while you are doing the project? *(C8, evaluate)*

Answer: The professional ethics required during the project included honesty, responsibility, proper documentation, teamwork, safe testing and correct reporting of results. The team had to report the system as a simulation and communication prototype rather than claiming it as a complete real-world fire-fighting system. All results, limitations and deviations had to be documented honestly. Ethical responsibility was also required while using external resources, images, models, research papers and software documentation. Such resources should be acknowledged properly in the report and presentation. During teamwork, each member's contribution had to be represented fairly. Safe handling of hardware was also an ethical responsibility because careless wiring or testing could damage equipment. The team also had to maintain professional communication with the project guide and teammates. Overall, the project required ethical behavior in technical development, result analysis, documentation, credit sharing and safety practices.

11. Do you think ethics and professionalism needs to be paid attention by students during study? If, yes, explain how it can be inculcated/introduced/implemented? *(C8, evaluate)*

Answer: Yes, ethics and professionalism need to be given attention by students during their study. Engineering work has direct impact on people, safety, environment and public trust. If students learn only technical implementation without understanding responsibility, they may ignore risks, copy work without credit, exaggerate results or neglect safety. Ethics can be inculcated through project-based learning, laboratory discipline, proper citation practice, teamwork evaluation and case studies of engineering failures. Students should be encouraged to document limitations honestly and to understand that incomplete or unsafe systems should not be presented as fully deployable solutions. Professionalism can also be developed through regular progress reviews, clear work division, meeting deadlines and presenting work in a structured manner. In this project, ethics and professionalism were practiced by maintaining clear contributions, using safe low-voltage hardware, documenting issues faced during testing and improving the system through continuous debugging.

12. Do you think environmental and sustainability limitations; ethical, health, safety, security, and risk issues; intellectual property; codes of practice and standards are sufficiently covered in the courses you have studied in your curriculum? *(C8, evaluate)*

Answer: Environmental and sustainability limitations, ethical issues, health and safety, security, risk, intellectual property and standards are covered to some extent in the engineering curriculum, but they can be strengthened further through practical project work. Courses usually introduce safety, environmental studies, professional ethics and basic standards in theory. However, students understand these topics more deeply when they apply them in real projects. In this project, the importance of these topics became clearer during hardware testing, serial communication, fire detection logic and simulation debugging. For example, a wrong coordinate, unsafe wiring or false fire confirmation can affect the reliability of the system. The curriculum provides the foundation, but more case studies, industrial examples, design reviews, risk-assessment exercises and documentation-based evaluation can improve student awareness. Therefore, these topics are present in the curriculum, but they should be more closely connected with laboratory work and final-year projects.

13. Have you gone through online classes, or a crash course in which you are familiarized with intellectual property rights as well as risk issues in professional environment? *(C8, evaluate)*

Answer: Yes, online learning resources, tutorials and documentation were referred to during the project to understand Unity, MATLAB serial communication, Arduino Nano usage and HC-05 Bluetooth communication. These resources helped in understanding software tools, communication workflow, debugging methods and project documentation. Awareness of intellectual property rights was developed by understanding that external assets, research papers, online code references and documentation must not be copied without acknowledgement. Risk issues in professional environments were also understood through general engineering safety practices, including safe handling of electronic components, avoiding high-risk experiments without supervision and clearly stating system limitations. Even if a formal crash course was not the main source, online learning and documentation played an important role in improving technical understanding. The project also encouraged self-learning because several issues such as serial port access, NavMesh placement, raycast blocking and truck positioning had to be solved through repeated testing.

14. In the beginning of your project did you evaluate environmental effects and sustainability factors in your work? *(C7, evaluate)*

Answer: Yes, environmental effects and sustainability factors were evaluated at the beginning of the project. Since the project deals with fire detection and emergency response, its environmental importance was clear from the initial stage. Fire incidents can damage buildings, release smoke, consume resources and create unsafe conditions. A faster detection and response system can help reduce such damage. At the same time, the team chose a simulation-based approach to avoid using real fire, smoke or hazardous materials during testing. This made the project environmentally safer and suitable for repeated demonstration. The hardware was limited to small low-power components, reducing resource use. Sustainability was also considered by designing the system as a modular prototype that can be improved in stages instead of building a large physical system immediately. The main limitation is that real UAV energy consumption and battery sustainability were not studied in the present phase.

15. Did you addressed the limitations of your project work and have you improved the results through continuous improvements in your project work? *(C5, create)*

Answer: Yes, the limitations of the project were identified and the results were improved through continuous testing and correction. During development, several problems were observed, including serial port access issues, coordinate parsing errors, drone teleportation, NavMesh placement errors, the drone passing through buildings, fire detection failure, raycast blocking by the navigation plane and incorrect truck height. Each issue was analyzed separately and corrected step by step. The drone movement was improved by using the DroneRoot structure and proper NavMeshAgent settings. Obstacle avoidance was improved using NavMesh and obstacle configuration. Fire detection was improved by combining trigger-zone detection with raycast confirmation and by using a separate fire raycast target. Truck deployment was corrected by using offsets and landing point arrangement. These continuous improvements helped the system reach a stable working condition. The final result successfully demonstrated coordinate reception, drone navigation, fire confirmation, truck deployment, drone landing and mission completion.

16. How did you plan your project, deadlines, maintaining dairy of each stage and improved the quality of the project? *(C14, understand)*

Answer: The project was planned by dividing the complete system into manageable stages. The first stage was literature study and understanding the project concept. The next stage was creating the Unity city environment and arranging the required objects such as drone, fire station, fire, drone pad and truck. After that, MATLAB communication, Arduino Nano and HC-05 Bluetooth pairing, and serial testing were carried out. The drone movement, NavMesh navigation, obstacle avoidance, fire detection and truck deployment were then implemented and tested. Deadlines were maintained by completing one module at a time and then integrating them. A project diary or progress record was maintained by noting the work completed, problems faced and corrections made. The quality of the project was improved through repeated testing, discussion among team members and debugging of each issue. Documentation and report preparation were carried out after the system workflow became stable.

17. Are you aware of the ethical clearance when you work in the field of health/medical applications.? *(C8, evaluate)*

Answer: Yes, awareness of ethical clearance is important when working in the field of health or medical applications. The present project is not a direct medical application and does not involve human subjects, patient data, clinical testing or medical devices. Therefore, formal medical ethical clearance was not required for this academic simulation. However, the project is related to public safety and emergency response, so ethical responsibility is still important. If the project is extended in the future to collect human data, monitor casualties, record images of people in emergency areas or support medical rescue decisions, ethical clearance and data privacy approval would become necessary. In such cases, informed consent, privacy protection, safe data handling, non-discrimination and responsible use of automated decision-making would need to be considered. The present work avoids these concerns by limiting itself to a simulated city environment and prototype communication.

18. Did you adopt any quantitative technique for any engineering activity related to your project? *(C3, evaluate)*

Answer: Yes, quantitative techniques were adopted in several engineering activities related to the project. Three-dimensional coordinate values were used to define the fire location and to move the drone toward the target. Distance-based calculations were used to check whether the drone reached the fire area, returned home or landed on the truck. Fixed numerical parameters such as take-off height, movement speed, stopping distance, detection radius, raycast distance and truck offset were used to tune the simulation. MATLAB received and displayed coordinate and mission status data, which supported mission tracking. The use of NavMeshAgent also involved quantitative settings such as speed, acceleration, angular speed and obstacle avoidance radius. These values were adjusted during testing to improve the behavior of the system. Therefore, quantitative methods were important for movement control, detection logic, debugging, response positioning and final mission validation.

19. What were the elements of your project work which addresses sustainable development and were you able to apply quantitative techniques to analyze and achieve your project goals? *(C7, evaluate)*

Answer: The project addresses sustainable development mainly through its application in early fire detection, reduction of emergency response delay and use of simulation-based testing. A faster fire verification and response workflow can reduce property damage, smoke release, resource loss and risk to human life. The project also supports sustainable engineering education because it allows repeated testing in Unity without creating real fire hazards or consuming large physical resources. Quantitative techniques were applied through coordinate-based movement, fixed flight height, detection distance, raycast length, truck offset and status tracking. These parameters helped analyze whether the project goals were achieved. For example, the drone had to reach the fire location, avoid obstacles, confirm the fire and deploy the truck. Although the project does not include real environmental sensors or field data, it demonstrates a sustainable prototype method by combining simulation, low-power communication hardware and modular software testing.

20. Did your project need the understanding of relevant legal requirements governing engineering activities you carried out as a part of your project work? Explain in detail. *(C8, evaluate)*

Answer: The project required awareness of relevant legal requirements at a basic level, although it was carried out as a simulation and laboratory prototype. Since no real UAV was flown outdoors, aviation rules and permissions were not directly applicable. However, if implemented physically, the project would need to follow drone operation regulations, local airspace rules, wireless communication regulations and fire safety norms. The HC-05 Bluetooth module was used only for short-range laboratory communication, so commercial communication licensing was not involved in the prototype. Legal requirements related to software licensing and asset usage were also relevant because Unity assets, 3D models and external references must be used according to their permitted terms. If the project is extended to real-world emergency response, data protection laws, safety certification, product liability and municipal fire response protocols would need to be considered. Thus, legal issues were limited in the present work but important for future expansion.

21. What are the legal, ethical practices you followed while working on project? *(C8, evaluate)*

Answer: The legal and ethical practices followed during the project included safe use of laboratory components, responsible documentation, proper acknowledgement of references and honest reporting of outcomes. The project was presented as a simulation and prototype communication model, not as a certified real-world fire response system. The team avoided using real fire or unsafe test conditions. External technical concepts, papers and tools were referred to for learning and were intended to be cited in the report. The work division between team members was documented honestly, including individual contributions and group contributions. The system limitations were also mentioned, such as the absence of real flame sensors, thermal cameras, GPS, real UAV control and physical fire suppression. These practices helped maintain academic integrity and professional responsibility. The team also followed safe electronic handling while working with Arduino Nano and HC-05 communication modules.

22. Are you sure that you abide IPR/copy right issues? *(C15, apply)*

Answer: Yes, the project work was carried out with awareness of IPR and copyright issues. The scripts and workflow developed for the project were prepared for academic use. Any external learning material, research papers, software documentation, images, assets or models used for understanding or demonstration should be acknowledged properly in the final report and presentation. The team did not claim ownership over general technologies such as Unity, MATLAB, Arduino, HC-05, NavMesh or UAV concepts. These were used as development tools and study references. If any third-party 3D models or graphical assets were used in the Unity scene, their license conditions should be checked before public sharing or publication. No patented method was knowingly copied or commercialized. Therefore, the project abides by IPR and copyright principles at the academic level, with the understanding that proper citations and asset credits must be maintained.

23. What online course you attended to improve your communication skills, Report writing, Oral presentation, Software used for writing report? *(C17, apply)*

Answer: Online resources and tool documentation were used to improve communication skills, report writing, presentation preparation and software usage. Unity documentation and tutorials were referred to understand NavMeshAgent, colliders, raycasting, object hierarchy and camera movement. MATLAB documentation was used to understand serialport communication, message formatting, reading and writing serial data. Arduino and HC-05 module resources were referred to understand Bluetooth pairing, serial testing and publisher-subscriber style communication. For report writing and presentation preparation, standard project report formats, IEEE reference style guidance and general academic writing practices were followed. Software tools used for writing and presentation included Microsoft Word for report preparation and Microsoft PowerPoint for presentation slides. These resources helped in improving technical communication, structuring the report and explaining the project workflow clearly.

24. In your project, was it needed to tackle risk issues, including health & safety, environmental and commercial risk, and of risk assessment and risk management techniques? Explain in detail. *(C5, create)*

Answer: Yes, the project required handling risk issues related to health and safety, environment, technical failure and demonstration reliability. Health and safety risks were reduced by avoiding real fire and using a simulation-based environment. Hardware risks were reduced by using low-voltage Arduino Nano and HC-05 modules and by checking serial connections carefully. Environmental risks were reduced because no physical fire, smoke or fuel was used. Technical risks included serial port errors, incorrect coordinate reception, drone movement failure, collision with buildings, false fire detection and truck positioning errors. These were managed through debugging, modular testing and parameter correction. Risk assessment was carried out informally by identifying what could go wrong in each module and correcting it before final demonstration. For example, the raycast was modified when the NavMesh plane blocked fire detection, and the truck offset was adjusted when the truck floated or went below ground. This improved the reliability of the final system.

25. How is the organization addressing a fire accident/human safety when working with machines? *(C9, evaluate)*

Answer: In an industrial or laboratory organization, fire accident and human safety are generally addressed through prevention, detection, emergency response and training. Fire extinguishers, emergency exits, alarm systems, electrical safety practices and proper storage of materials are important safety measures. When working with machines, operators are expected to follow standard operating procedures, wear necessary personal protective equipment, avoid unsafe wiring and switch off equipment during maintenance. In the context of this project, the organization-level safety idea is represented through the simulated emergency response workflow. The system receives a fire coordinate, sends a drone for verification and deploys a response vehicle only after confirmation. This reduces unnecessary human exposure to hazardous zones. The project also demonstrates how automation and remote verification can support safer decision-making during fire incidents. In real implementation, it would need integration with fire alarm systems, control rooms and trained emergency personnel.

26. Process of teamwork. How each of you are involved in the team? What part the work is addressed by you.? *(C16, evaluate)*

Answer: The project followed a teamwork process in which tasks were divided according to the strengths and responsibilities of each member. Spandan Kumar contributed mainly to the Unity part of the project. This included development of the Unity city environment, setting up the DroneRoot structure, arranging the drone model, implementing NavMesh navigation, testing drone movement, correcting obstacle avoidance and carrying out final simulation testing. H Anupam Nair contributed mainly to MATLAB serial communication, FIRE coordinate publishing, status receiving loop and mission report display. Both team members worked on the Arduino Nano and HC-05 Bluetooth publisher-subscriber circuit, including pairing, serial testing and communication validation. Both members also contributed to documentation, literature study, testing support, result analysis and report preparation. The teamwork process involved discussion, division of modules, integration of individual work, identification of errors and repeated testing until the final workflow functioned correctly.

27. Have you filed patent, IPR, or published your work? Give more details. *(C17, evaluate)*

Answer: No patent, IPR filing or publication has been completed for this project at the current stage. The project has been developed as an academic in-house project for demonstration and learning purposes. The present work integrates known technologies such as Unity simulation, MATLAB serial communication, Arduino Nano, HC-05 Bluetooth modules, NavMesh navigation and raycast-based detection into a working prototype. Since the project is still at simulation and prototype level, further development would be required before considering a patent or publication. If future work adds real UAV hardware, advanced fire sensing, image processing, GPS navigation, IoT integration or a novel detection algorithm, the team may explore publication in a conference or journal. Any such future publication would include proper references, contribution details, experimental validation and comparison with existing work. For now, the documentation and report represent the formal academic record of the project.

28. How you documented the literature review, your analysis on their results, discussion with the supervisor and team members, provide the documents on weekly basis. Put as one chapter in final report. *(C4, evaluate).*

Answer: The literature review was documented by collecting research papers and technical sources related to UAV fire monitoring, fire detection, path planning, communication systems, simulation tools and embedded systems. The important findings from these sources were summarized and connected to the project requirements. The analysis focused on how existing works support different parts of the project, such as drone-based monitoring, image or sensor-based detection, path planning and serial communication. Discussions with the supervisor and team members helped refine the scope and methodology. Weekly progress was documented through project notes, screenshots, code updates, testing observations and corrections made during debugging. These records were later organized into the final report as chapters such as introduction, literature review and objectives, methodology, results and discussion, contribution of students, conclusion and future scope. This documentation process helped show both the technical development and the learning progress of the team.

29. Have you sensitized about inclusion and diversity in the team? If yes, what are the diversification in the team in terms of religion, gender, ethnicity, etc? *(C11, apply).*

Answer: Yes, the team was sensitized about inclusion and diversity during the project work. The project was completed through cooperation between team members, with responsibilities divided according to work requirements and technical involvement rather than personal background. The focus was on fair participation, respectful communication, equal opportunity to contribute and proper recognition of each member's work. The team avoided discrimination based on gender, religion, ethnicity, language or personal identity. Since the team consisted of Spandan Kumar and H Anupam Nair, the diversification was mainly in terms of individual skills, technical interests and assigned project responsibilities. One member focused strongly on Unity simulation and navigation, while the other focused strongly on MATLAB communication and status handling. Both members contributed together to the hardware communication circuit, testing and documentation. This diversity in skills improved the quality of the project.

30. How were you able to keep yourself updated with the technology? How you incorporated advanced technology in your project. *(C18, lifelong learning)*

Answer: The team kept updated with technology by referring to tool documentation, tutorials, research literature and repeated practical testing. Unity features such as NavMeshAgent, object hierarchy, colliders, raycasting and camera control were studied and incorporated into the project. MATLAB serial communication functions were used to send coordinates and receive mission status. Arduino Nano and HC-05 Bluetooth modules were used to demonstrate embedded wireless communication through a publisher-subscriber model. The project incorporated advanced technology concepts such as UAV-based emergency response, simulation-driven design, autonomous navigation, obstacle avoidance, hybrid sensing and cyber-physical system integration. The system was not limited to animation; it combined software simulation, embedded communication and mission logic. Continuous debugging also helped the team learn new technical solutions, such as avoiding NavMesh-related errors, correcting raycast blocking and stabilizing drone movement through a root-object structure.

31. Which are the laboratory skills you found applicable to your project? Explain. *(C12, apply)*

Answer: The laboratory skills applicable to the project included circuit connection, serial communication testing, Bluetooth module pairing, software installation, simulation setup, debugging and result observation. In the hardware part, Arduino Nano and HC-05 modules required proper wiring, power connection, pairing and communication validation. Serial monitor testing and port selection were important to check whether data was transmitted correctly. In the software part, Unity laboratory skills included importing and arranging models, creating GameObjects, assigning scripts, setting colliders, configuring NavMesh surfaces, tuning NavMeshAgent parameters and checking console logs. MATLAB skills included writing and running scripts, configuring the serial port, formatting coordinate messages and reading status updates. Debugging was one of the most important laboratory skills because the project required solving errors related to COM port access, parsing, navigation, detection and object placement. These skills together supported successful completion of the project.

Annexure 4
Project Work (In-house) Classification

Student Name: Spandan Kumar / H Anupam Nair
Registration No: 220929337 / 220929228

Table 1: Classification based on project domain classification

Domain	✓ Tick
Product	✓
Application	✓
Review	
Research	
Management	

Table 2: classification based on societal consideration

Societal Impact	✓ Tick
Ethics	✓
Safety	✓
Environmental	✓
Commercial	
Economical	✓
Social	✓

Signature of Student: **Name and Signature of Supervisor:**

Spandan Kumar:

H Anupam Nair:

Date:

Annexure 5

Project Details

Student Details			
Student Name	**Spandan Kumar**		
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Project Work (In-house) Details			
Project Title	**Wildfire Spread Prediction and live Drone-Based Visual Feedback System**		
Date of Synopsis	27 January 2026	Date of Final Presentation	19 May 2026

Internal Supervisor Details, If any Co-supervisor	
Name of the Supervisor	**Dr. Shweta Vincent**
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Student Details			
Student Name	**H Anupam Nair**		
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Project Work (In-house) Details			
Project Title	**Wildfire Spread Prediction and live Drone-Based Visual Feedback System**		
Date of Synopsis	27 January 2026	Date of Final Presentation	19 May 2026

Internal Supervisor Details, If any Co-supervisor	
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